Lecture #9: Machine learning for molecular simulation

Slides by Artem Dembitskiy

Previously on

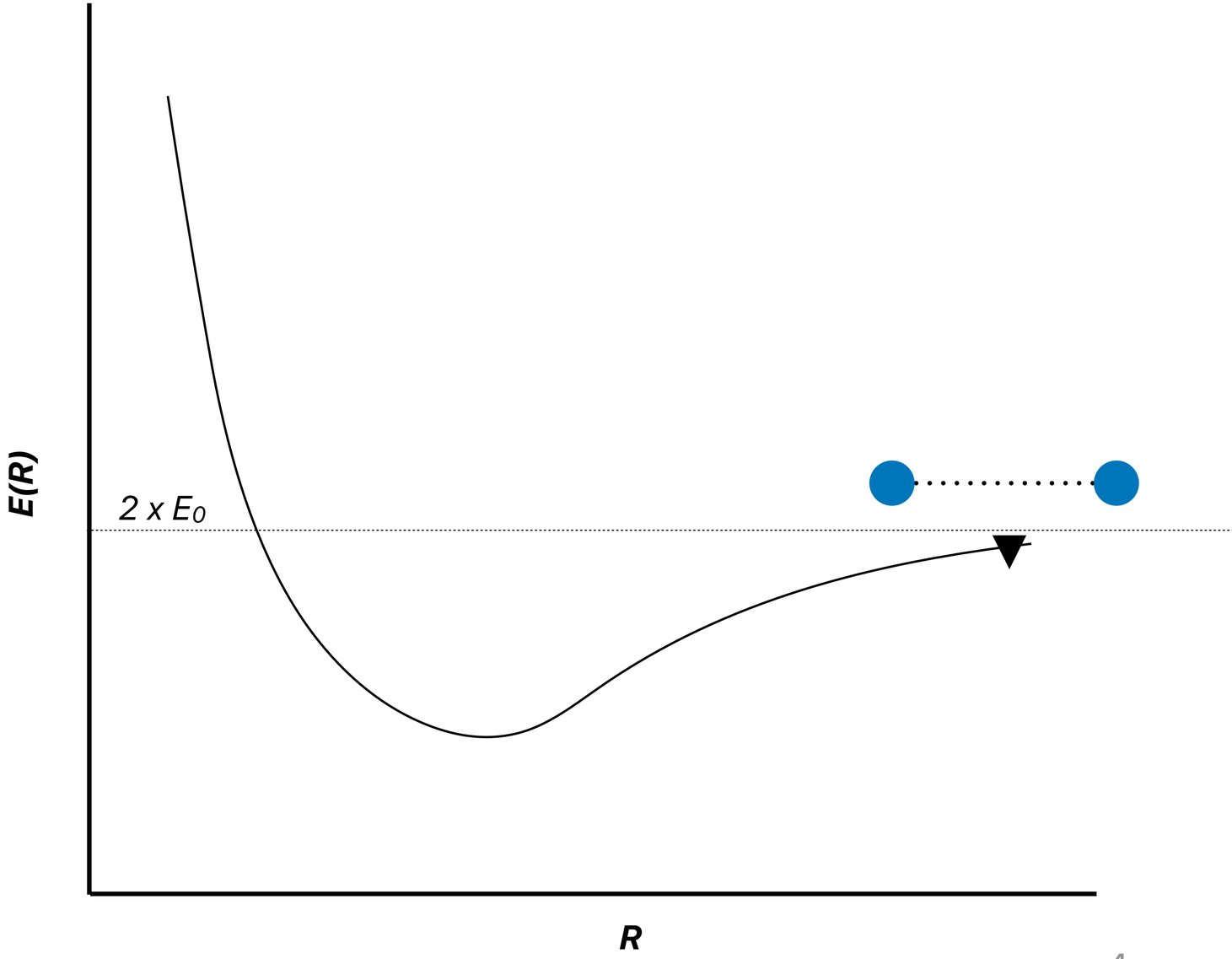
- Graph neural networks
- Message passing
- Permutation invariance

Goals/Agenda

- Potential energy surface on H_2 molecule example
- Molecular dynamics basics
- Interatomic potential
  - Energy and forces
- ML interatomic potentials (MLIPs)
  - Locality principle
  - Generating data
  - Fitting and validation
  - Active learning
  - Invariant, equivariant, and unconstrained models
- Universal MLIPs
  - Datasets, models, benchmarks

Consider two isolated H atoms at the infinite distance from each other.

- The total potential energy of this system is 2 x E_0
- Let's iteratively move these atoms closer to each other and measure potential energy of the system
- As a result we will obtain a relationship E_i vs. R_i



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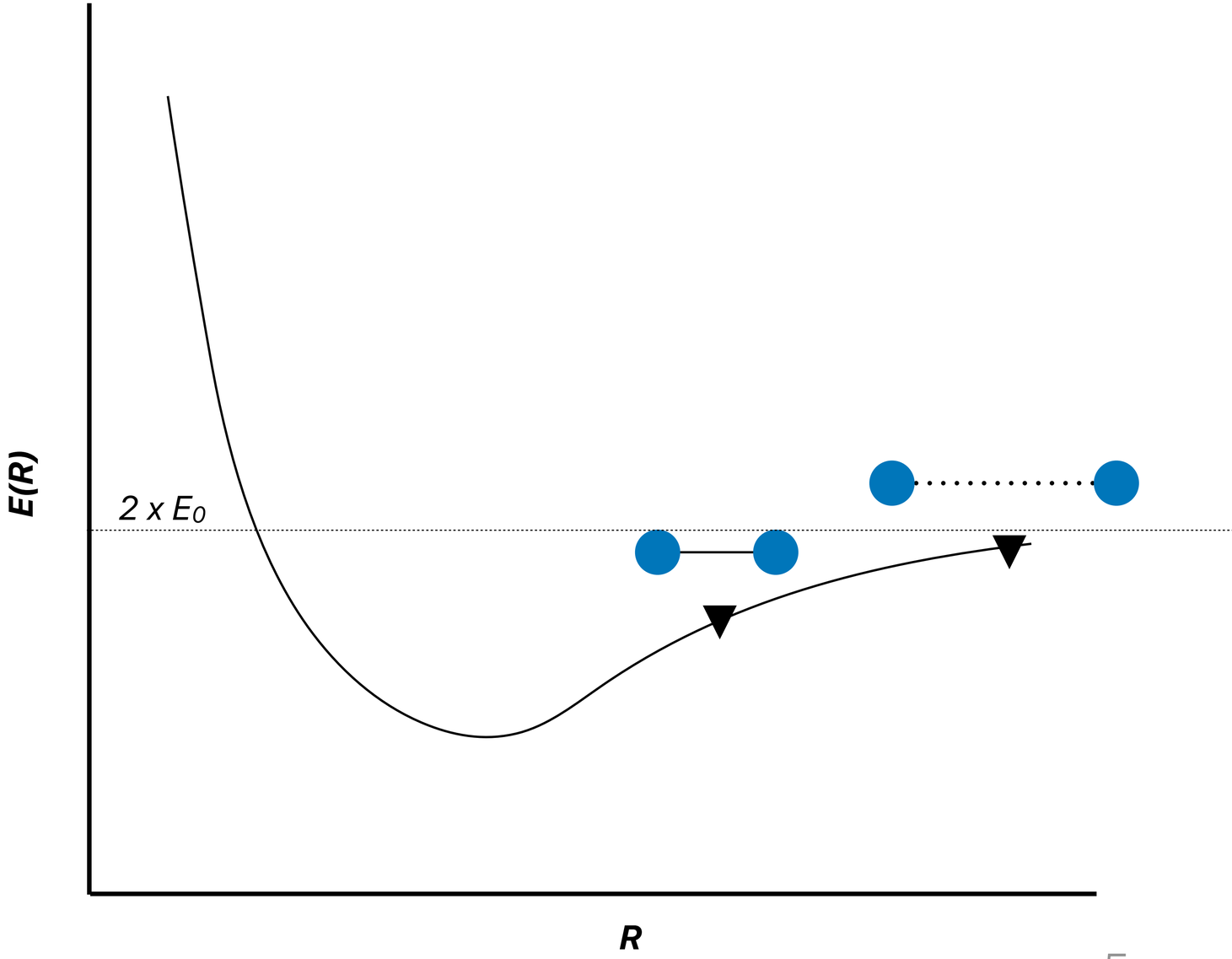

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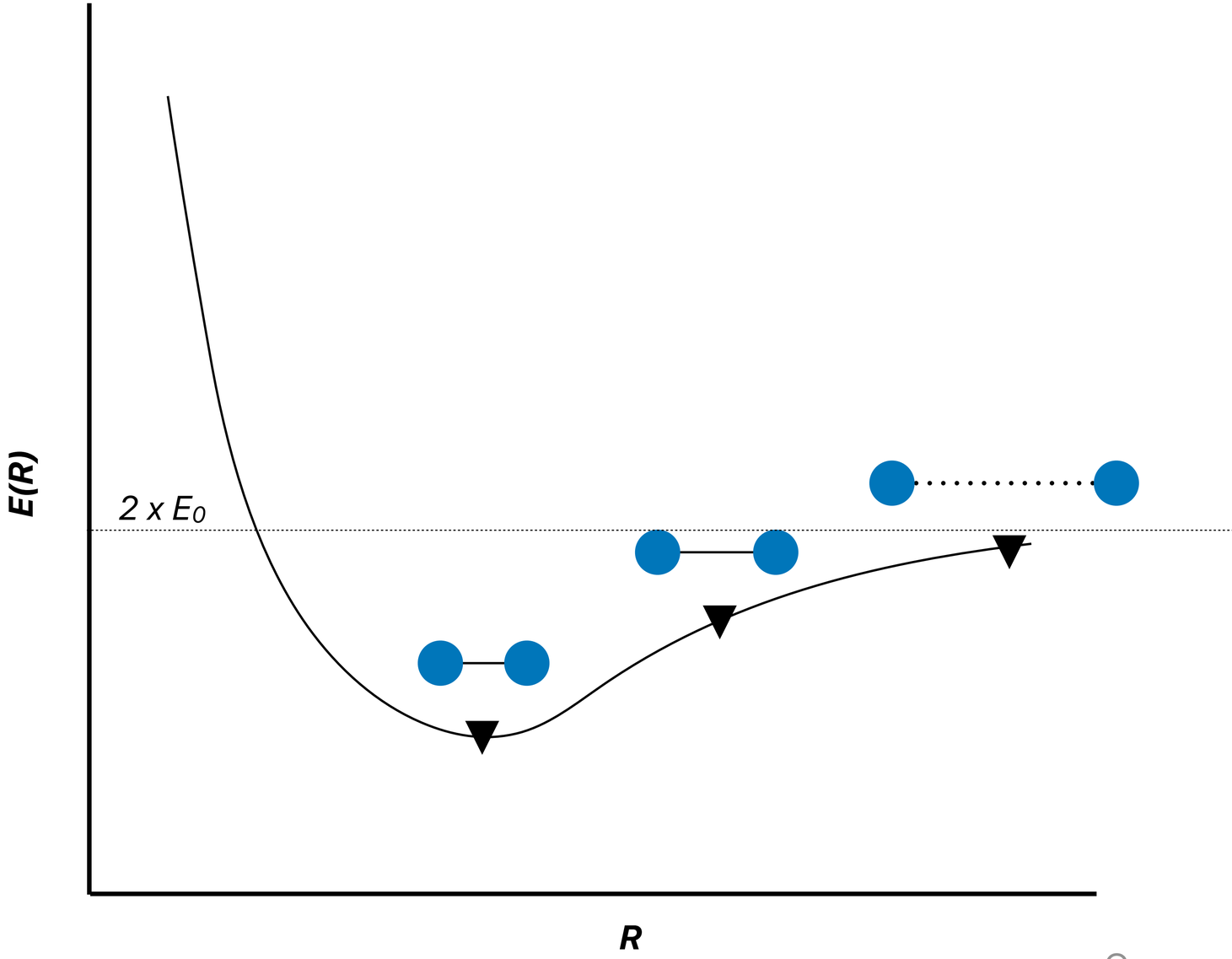

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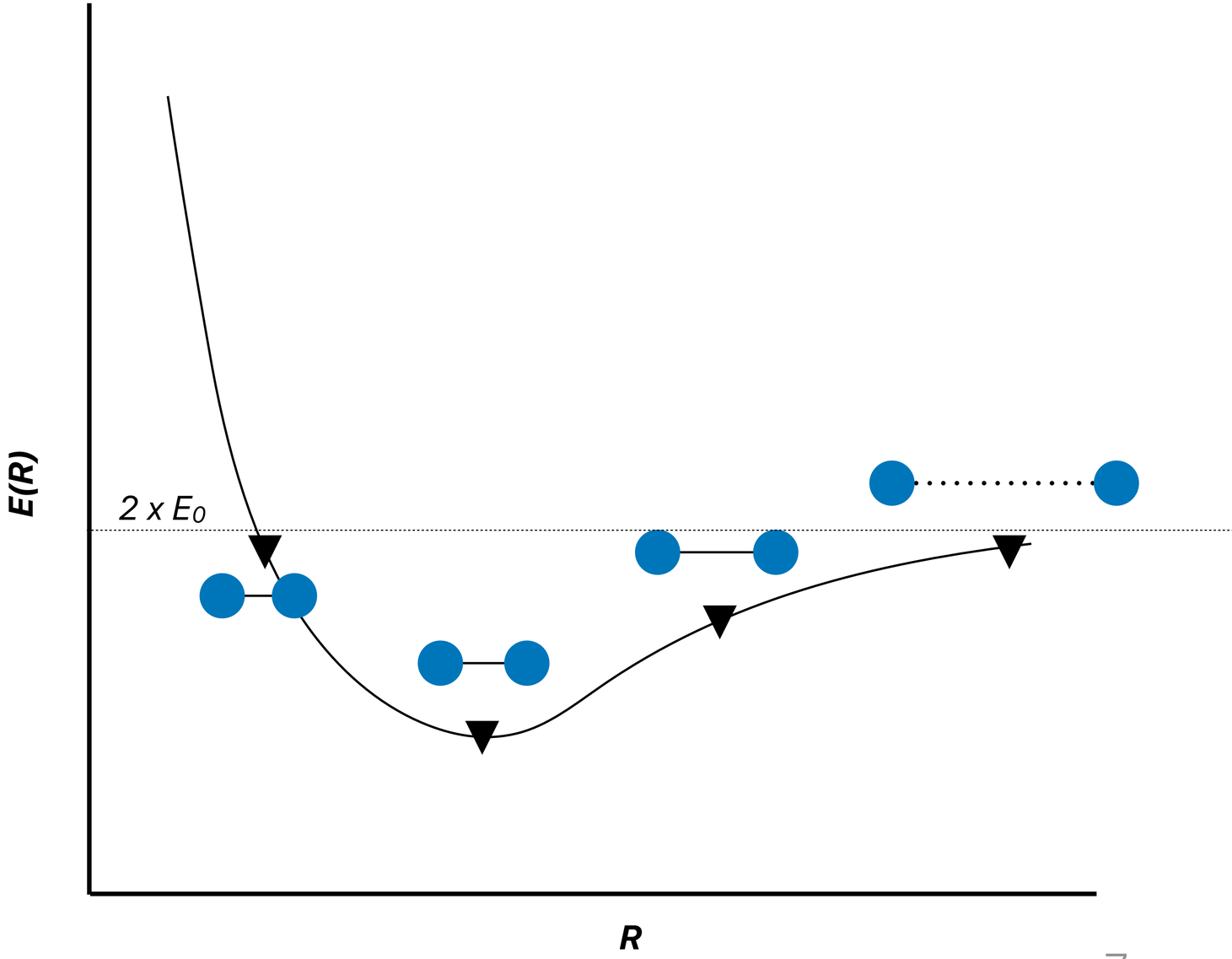
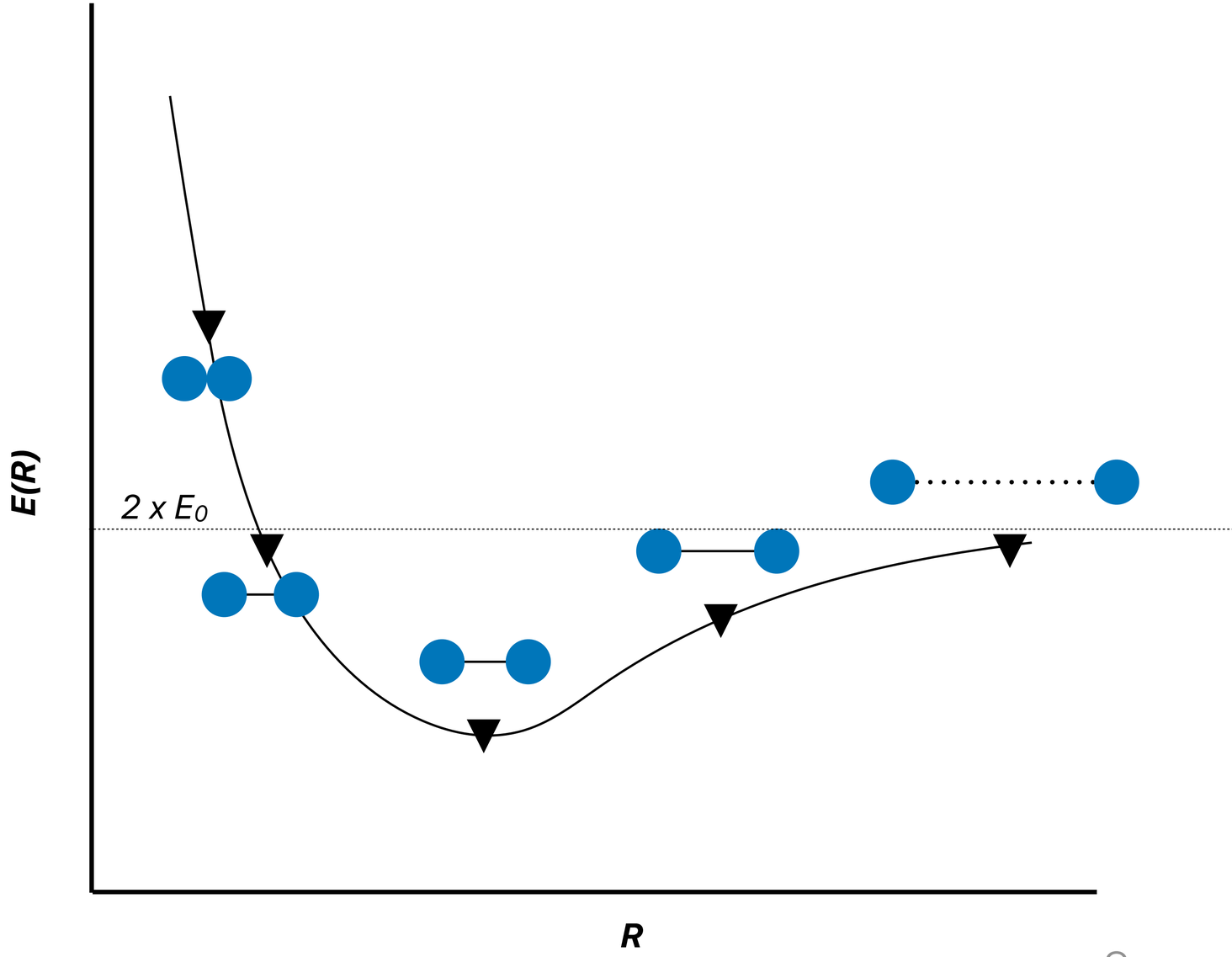


Potential energy surface

- The relationship between the total potential energy of a system and its geometry is called a potential energy surface (PES)



- More complex systems have more complex PES.
- In principle, $E = E(\mathbf{R})$ depends on $3N - 6$ degrees of freedom, where N is the number of atoms.
- To simplify analysis, scientists reduce the number of degrees of freedom.
- Example: rotating one molecular fragment relative to another gives a 1D PES: $E = E$ (rotation angle).

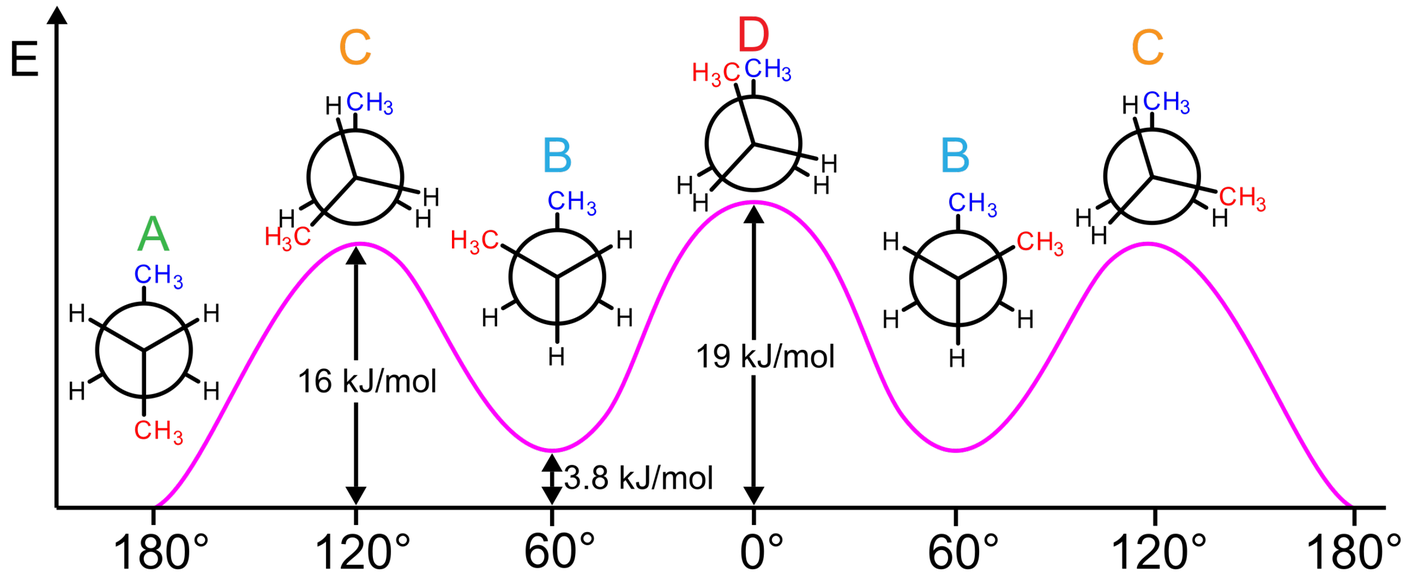
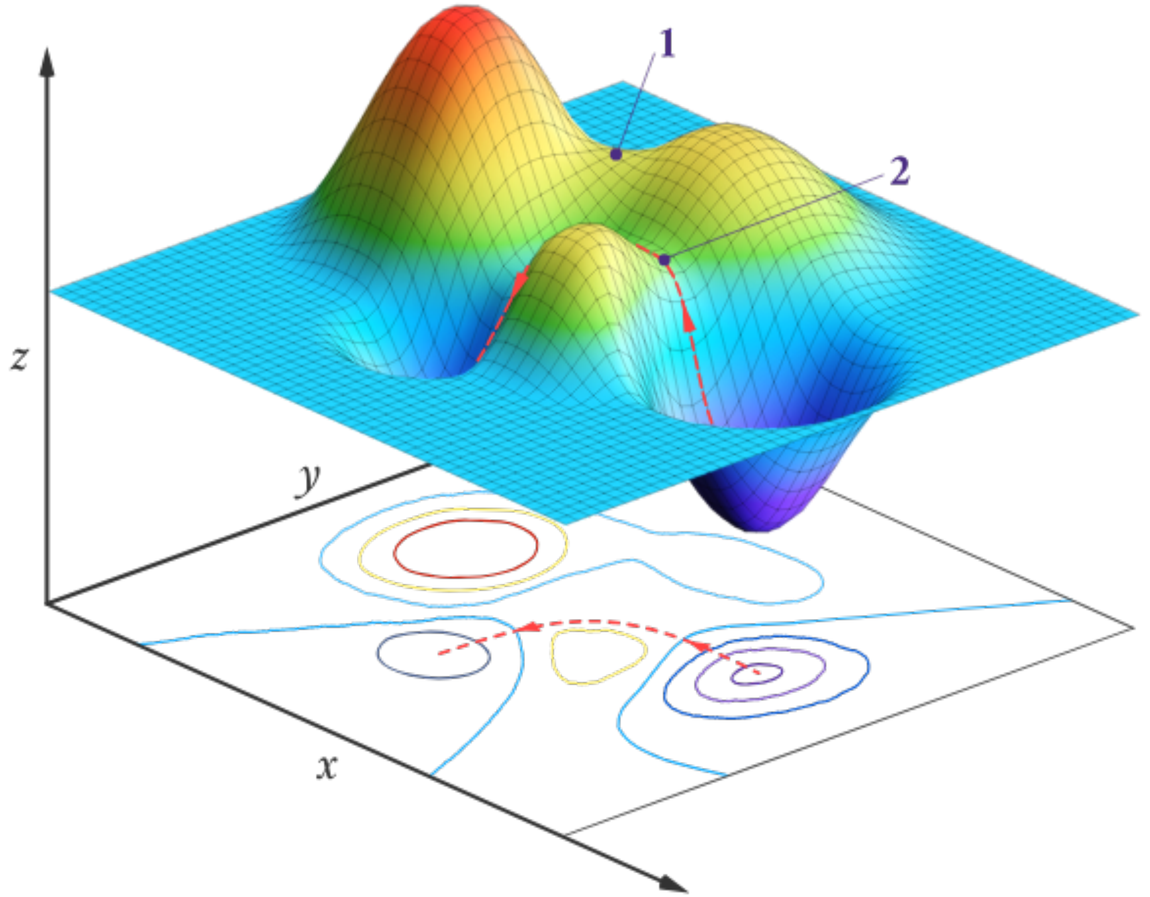


Image source

What shape PES tells you about?

- Global minimum (stable state)
- Local metastable states
- Transition states
- Reaction pathways
- Vibrational frequencies
- Forces
- ...

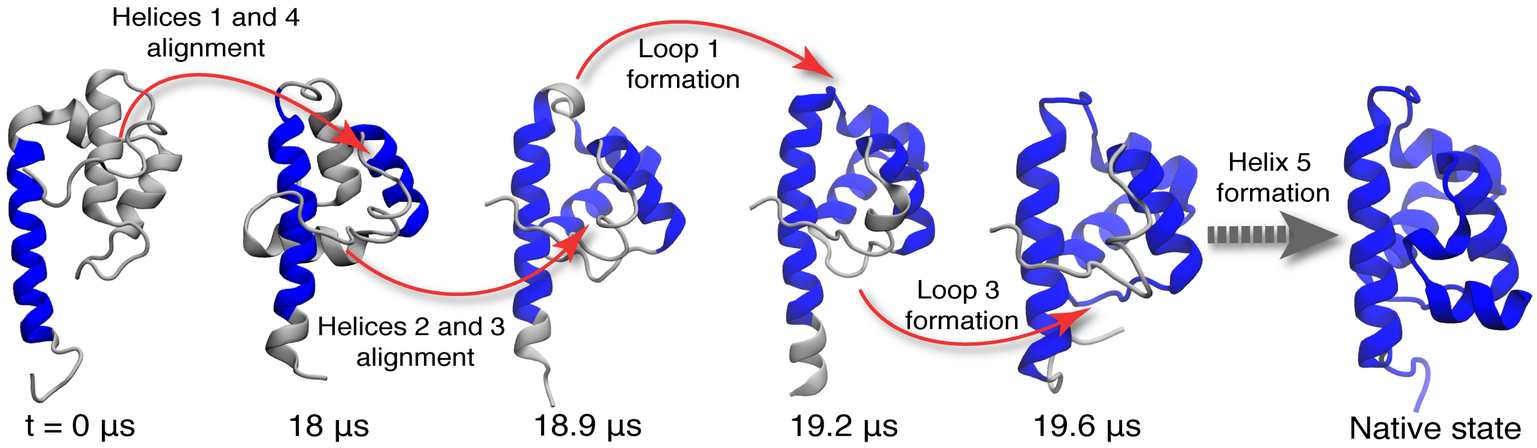


But we don’t know PES shape in advance. We have to sample it.

Molecular dynamics

...is used to study the dynamics of atomic systems

- i.e. evolution at finite temperature
- i.e., sampling PES



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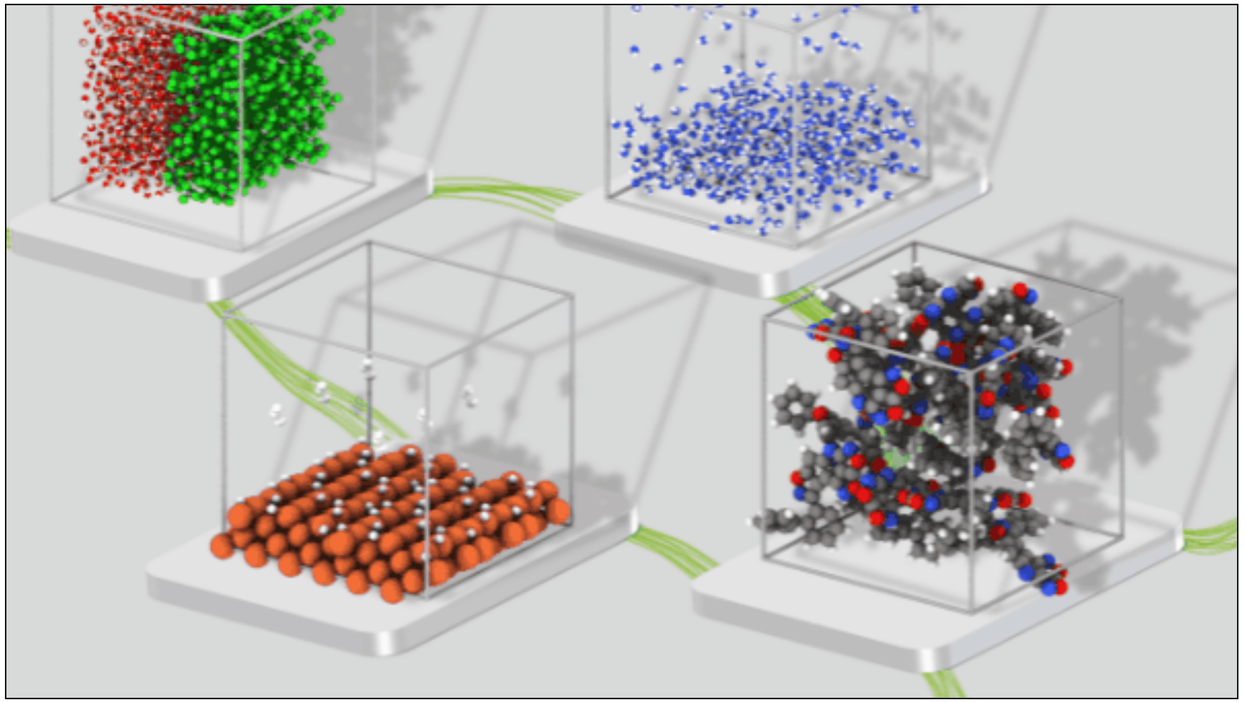
Applications of molecular dynamics (MD) analysis

- Configurational space:
  - E.g., geometry of molecular conformers
- Time-averaged observables
  - <V(T)>, radial distribution functions, average bond lengths, etc.
- Rate of some events
  - Ionic jumps, defects formation, etc
- Study of any imaginable process at atomistic scale
  - Crystallization, melting
  - Interaction and evolution of molecules
  - Catalyst surface reconstruction
  - Nano indentation



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Basics of molecular dynamics

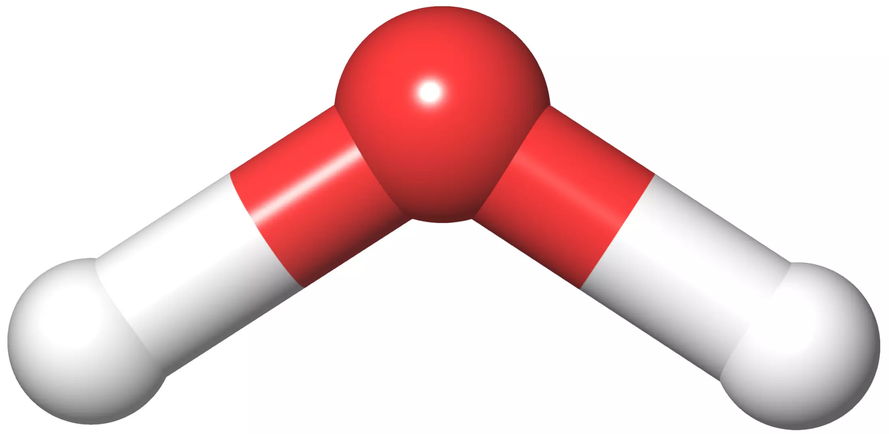
The net force acting on particle is directly proportional to its acceleration:

$$F_i = m_i \frac{\partial^2 r_i}{\partial t^2}$$

At the same time:

$F_i = -\frac{\partial E}{\partial r_i}$, where E is PES, $E(R)$

$R = R(r_1, \ldots, r_i)$ - is the state of the system, i.e. its geometry



Integrating Newton's laws of motion

- We want to find $r_i(t)$, i.e. $R(t)$

– We **can't** find it analytically for more than two interacting particles (see three-body problem)

- We **can** solve it numerically

Finite difference method

The shape of $E(R)$ is known at the given moment t

The acceleration for each particle is given by:

$$a_i = \frac{F_i}{m_i} = -\frac{1}{m_i}\frac{\partial E}{\partial r_i} \text{ eq. (1)}$$

1) Initialize velocities
2) Select a small timestep δt (it should correctly describe the changes in your system)
3) Calculate forces
4) Integrate eq. (1) within $(t, t + \delta t)$ window to get velocities $\boldsymbol{v_i}(t + \delta t)$
5) Update coordinates using calculated velocities for each particle $\boldsymbol{r_i} = \boldsymbol{r_i} + \delta t \boldsymbol{v_i}$
6) Repeat 2-4 steps ***N*** times
7) Analyze the discrete trajectory obtained

MD scheme

Atomic positions $\boldsymbol{R_i}$

↓

Calculate $\boldsymbol{E(R_i)}$

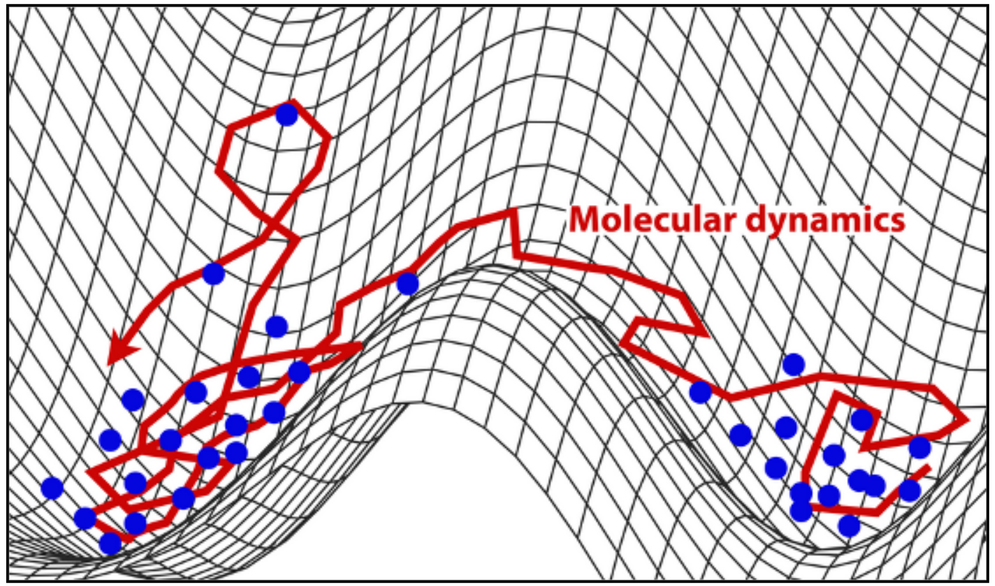
↓

Force on nuclei
$\boldsymbol{F}$ = -grad $\boldsymbol{E(R_i)}$

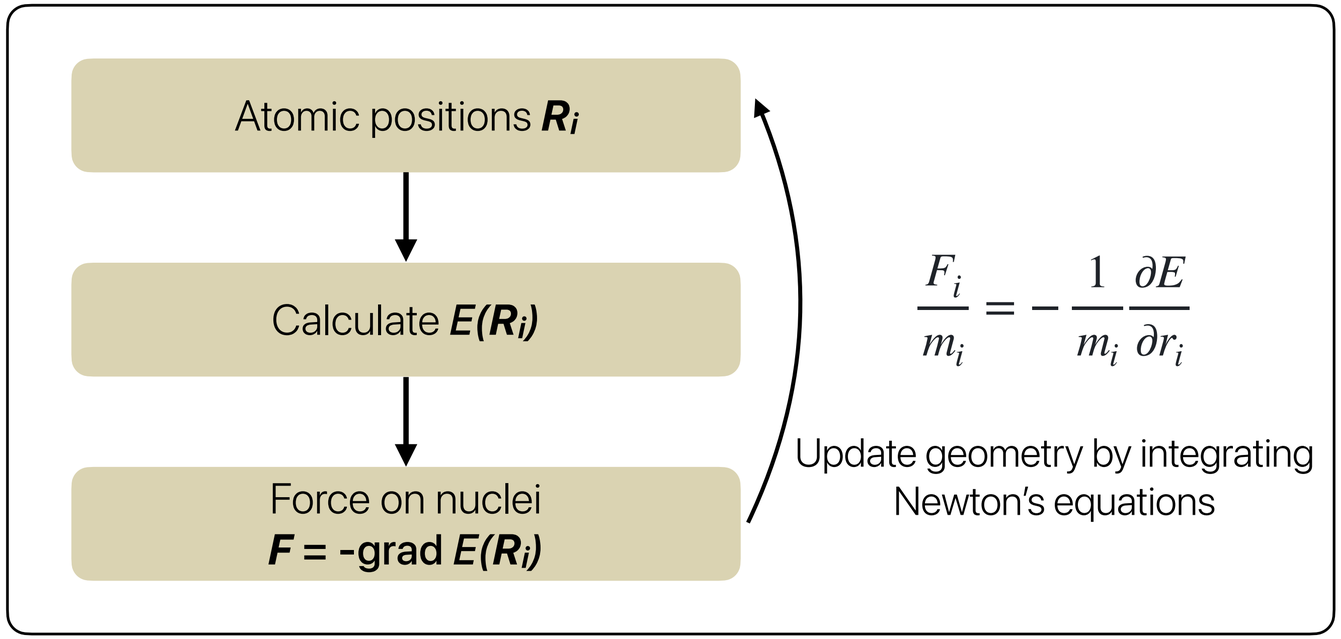
$$\frac{F_i}{m_i} = -\frac{1}{m_i}\frac{\partial E}{\partial r_i}$$

Update geometry by integrating Newton's equations

Sampled discrete trajectory



MD scheme



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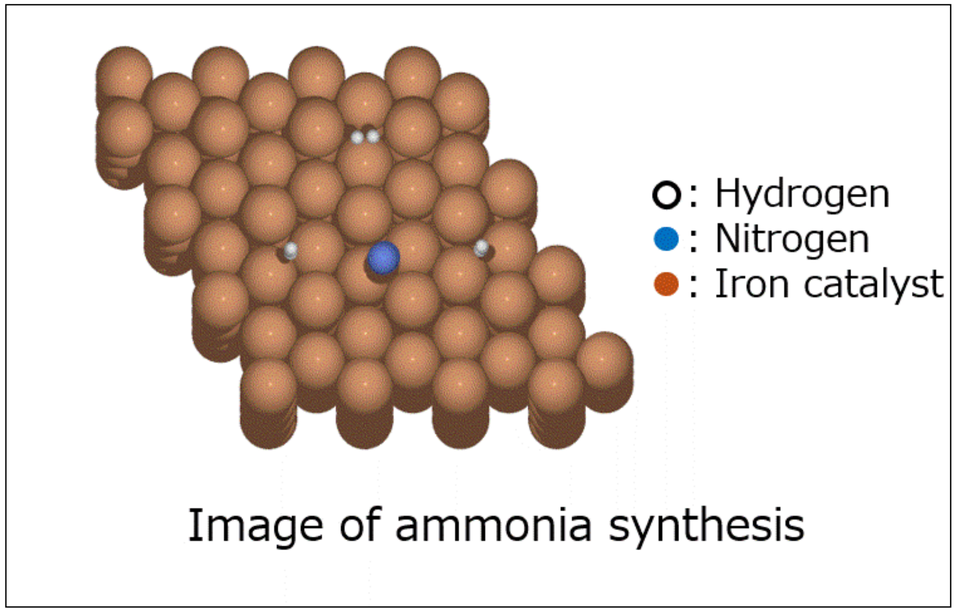
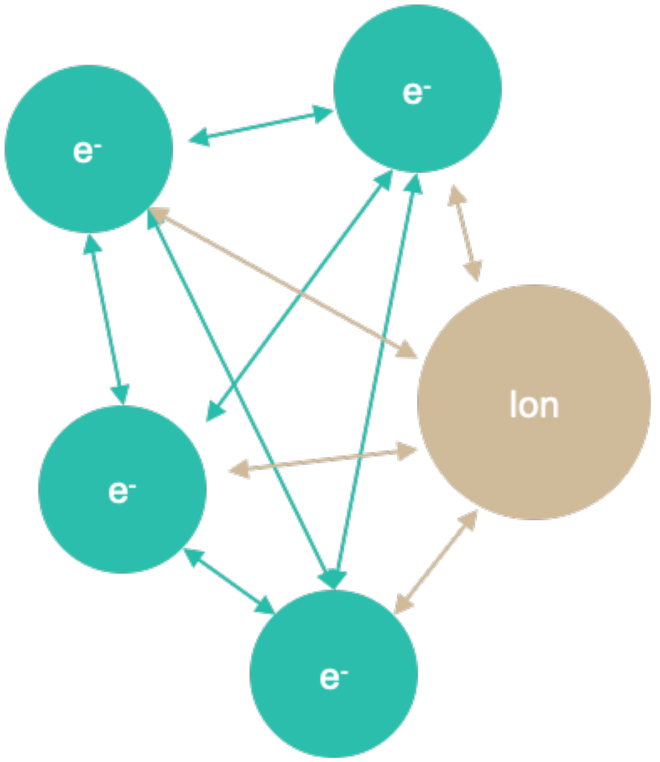


Gif source

How to calculate forces and energy?

Quantum mechanics says that:

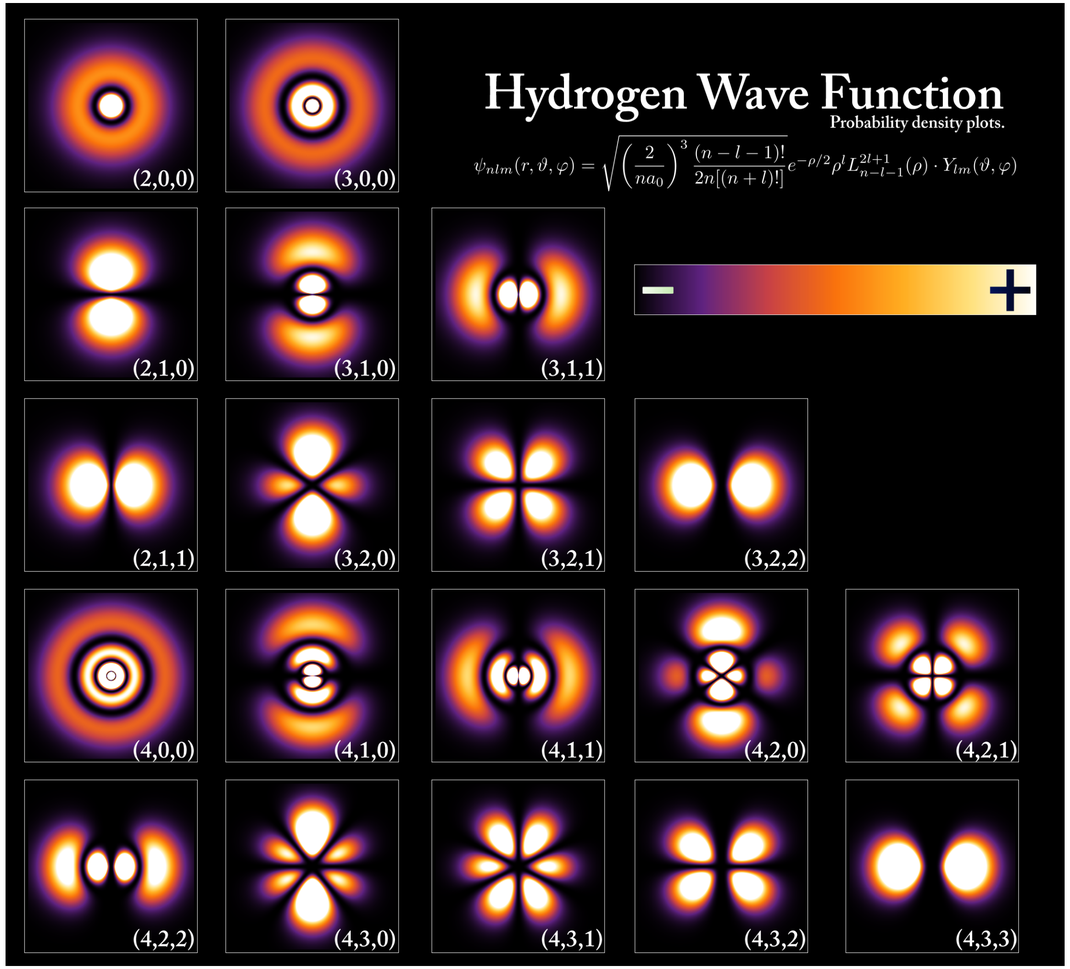
Solving Schrödinger equation gives exact description of a molecular system



$$\hat{H}\Psi(\mathbf{r}_1, \mathbf{r}_2, \ldots, \mathbf{r}_N) = E\Psi(\mathbf{r}_1, \mathbf{r}_2, \ldots, \mathbf{r}_N)$$

However, exact solution is possible only for a very simple models

- Schrödinger equation can be exactly solved only for hydrogen-like systems
- Approximation is needed for modeling more complex systems!



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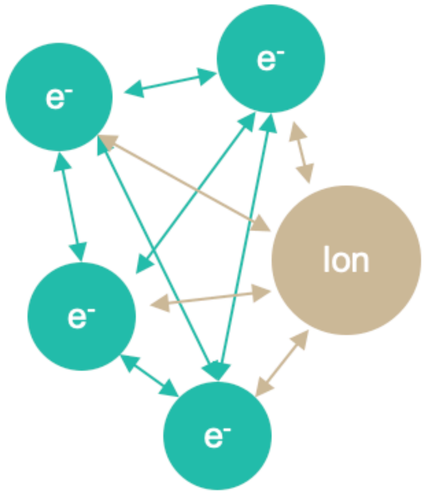
https://en.wikipedia.org/wiki/Atomic_orbital

Density functional theory approximation

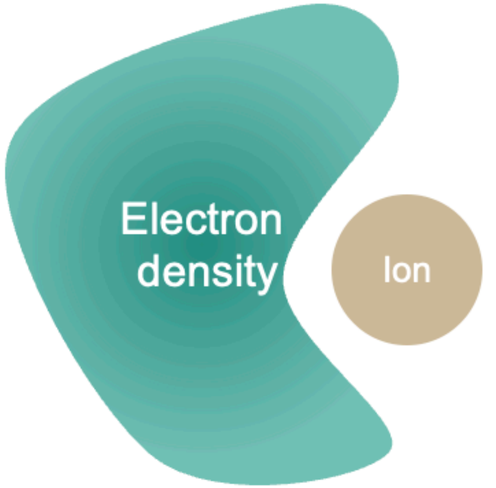
- Many-body wavefunction -> electronic density
- Interacting electrons -> non-interacting particles
- Many-particle interactions are included in effective exchange-correlation functional
- Ground state of a system is a functional of the electronic density — *E = E[n(**r**)]*, ***r*** - electronic coordinates

Many-body problem



$$\Psi = \Psi(\mathbf{r}_1, \mathbf{r}_2, \ldots, \mathbf{r}_N)$$

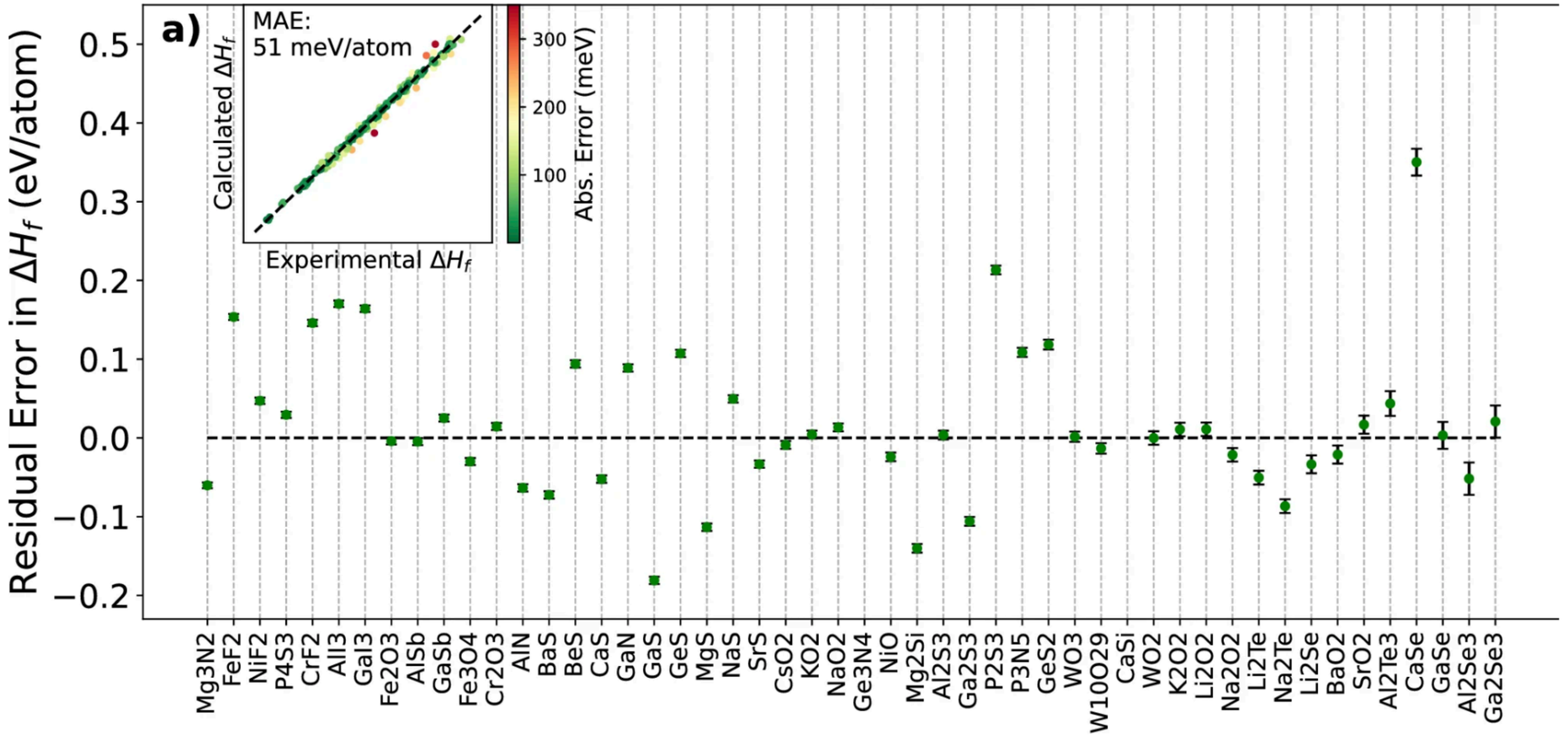
DFT perspective



$$n(\mathbf{r}) = \sum_{i=1}^{N} |\phi_i(\mathbf{r})|^2$$

Figure redrawn from MRS Bulletin 36, 169–174 (2011)

DFT vs. Experiment



Limitations for MD simulations using DFT

The potential energy surface is calculated using quantum chemistry method

- Pros
  - “Exact”
  - Yields electronic structure
- Cons
  - O(N^3) scaling law, N - number of electrons
  - Small systems can be studied (~100 atoms) - **very expensive**
    - Poor statistics
    - Artificial size-effects for periodic systems
  - Short time scales (~100 ps)
    - Poor statistics

Approximate evaluation of time required

- It takes ~60 seconds to do a single MD step for a system with ~100 atoms on a 64 CPU cores
- 2 fs timestep and 100 ps trajectory -> 100 * 1000 / 2 -> 50,000 MD steps -> **1 month**
- To study 1000 atoms system it would take **x 10^3 more time!**

Machine learning interatomic potentials (MLIPs) as a substitute for DFT

We want to move from quantum mechanics:

$$H(\{Z, \mathbf{R}\}) \xrightarrow{\Psi} E$$

To some surrogate model:

$$F(\{Z, \mathbf{R}\}, W) \xrightarrow{\mathbf{W}} E*$$

Where F is a learnable function, e.g.:

$$\min_{W} \sum_{i} (E_i - E_i^*)^2$$

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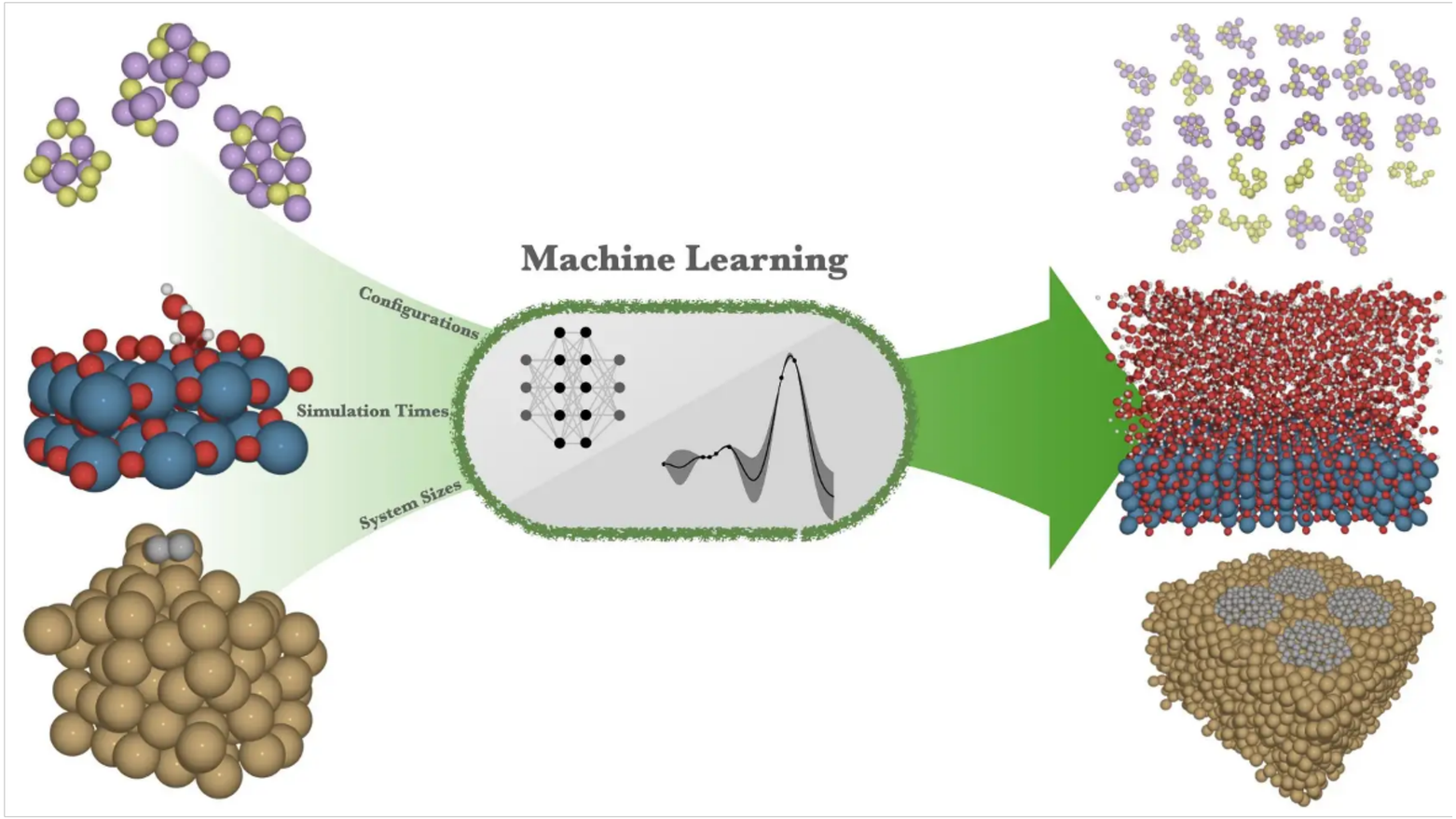


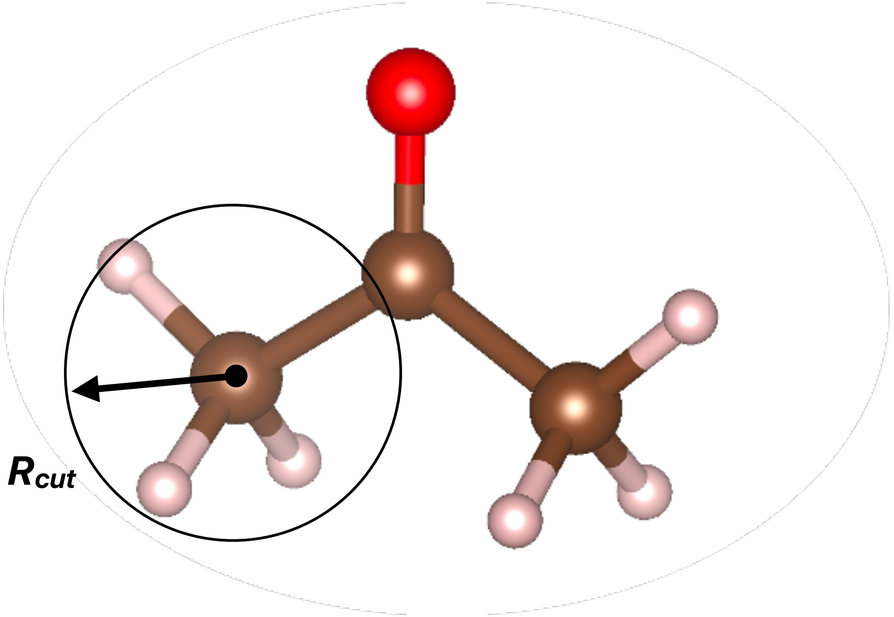
Image source

Assumptions for constructing MLIPs

- Atoms — indivisible point-like particles
  - No electronic structure!
- Locality principle - enables O(N) scaling law,

  N - number of atoms
  - The potential energy is calculated as the sum of atomic potential energies (site energies)

$$E_{pot} = \sum_{i \in N_{atoms}} E_{i,atomic}$$



$$E_{i,atomic} = \phi(\{\vec{r}_j\}_i, \{Z_j\}_i)$$

A learnable function mapping local atomic environment to the atomic energy

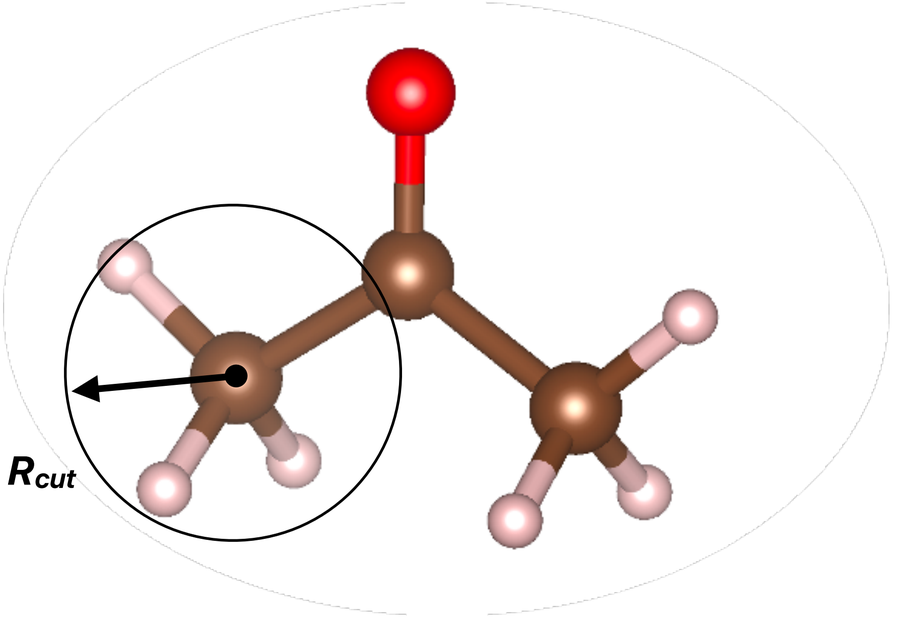
Requirements for MLIP

- If we rotate, translate, or invert coordinate system -> properties must transform accordingly

- Invariance to permutation of atomic indices:

$$E(r_1, r_2 \ldots, r_N) = E(r_2, r_1, \ldots, r_N)$$

- Near-DFT accuracy at a fraction of a cost
  - Otherwise, what's the point?



$$E_{i,atomic} = \phi(\{\vec{r}_j\}_i, \{Z_j\}_i)$$

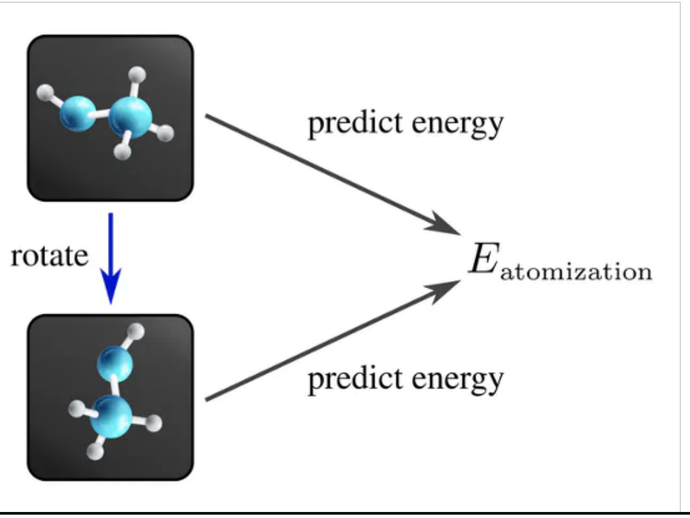
A learnable function mapping local atomic environment to the atomic energy

What does "transform accordingly" mean?

- There are invariant and equivariant physical properties
- Scalar values (e.g., energy, mass) must not change under E(3) symmetry group
- Tensorial properties (e.g., forces, dielectric tensor) must change predictably

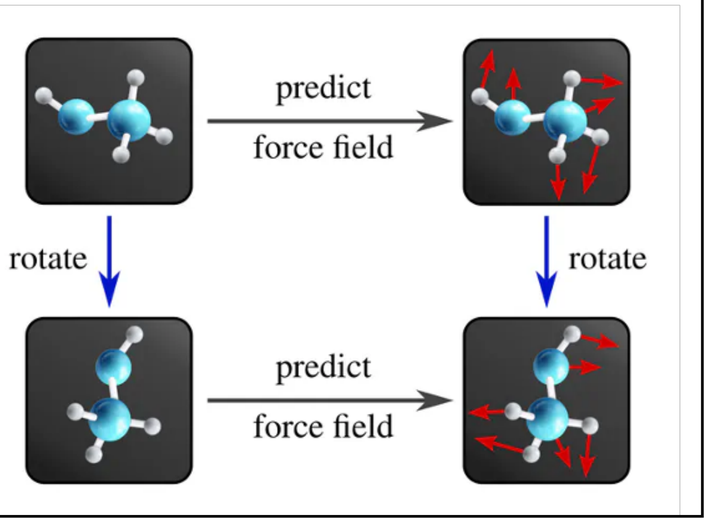
Invariance

$$f(g \triangleright_X x) = f(x) \qquad \forall\, g \in G,\ x \in X,$$



Equivariance

$$f(g \triangleright_X x) = g \triangleright_Y f(x) \qquad \forall\, g \in G,\ x \in X$$



$$\mathbf{x}' = R\mathbf{x} + \mathbf{t}$$

Image from: https://maurice-weiler.gitlab.io/blog_post/cnn-book_1_equivariant_networks/

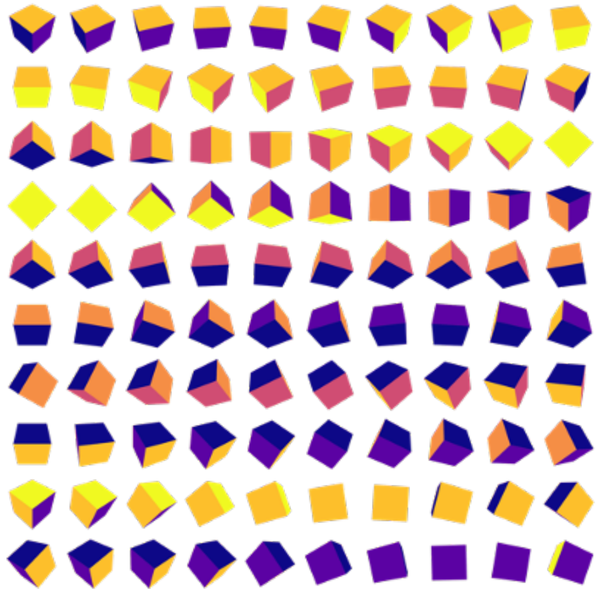
Physical bias vs. Augmentations

- We can enforce invariance/equivarince by using data representations and transformation operations satisfying E(3) symmetry

- Or we can augment data and hope the model will learn the symmetry (aka unconstrained models)

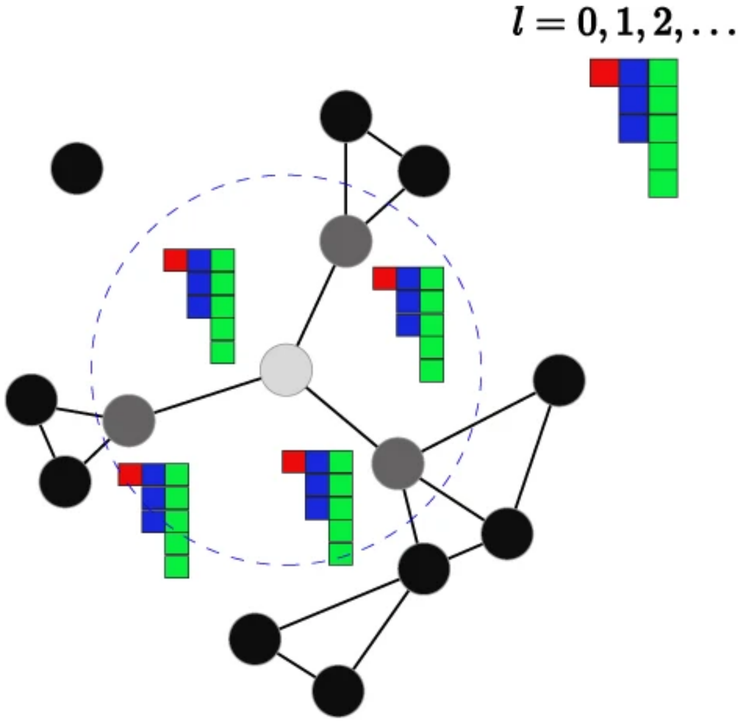
Training with symmetry

Learning symmetry



Examples

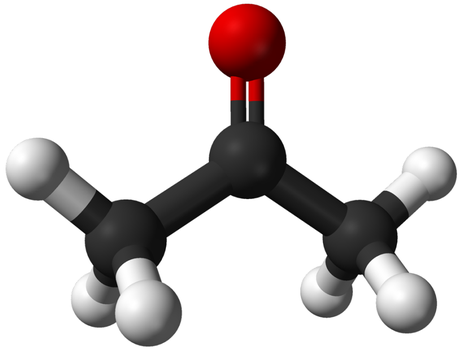
- ML potentials
  - Gaussian approximation potential (GAP)
  - Atomic cluster expansion (ACE)
  - Smooth overlap of atomic positions (SOAP)
  - Moment tensor potentials (MTP)
- NNs
  - The Behler-Parrinello Neural Network <- invariant
- GNNs
  - M3GNet <- invariant
  - MACE <- equivariant
  - Nequip/Allegro <- equivariant
  - Point edge transformer (PET) <- unconstrained

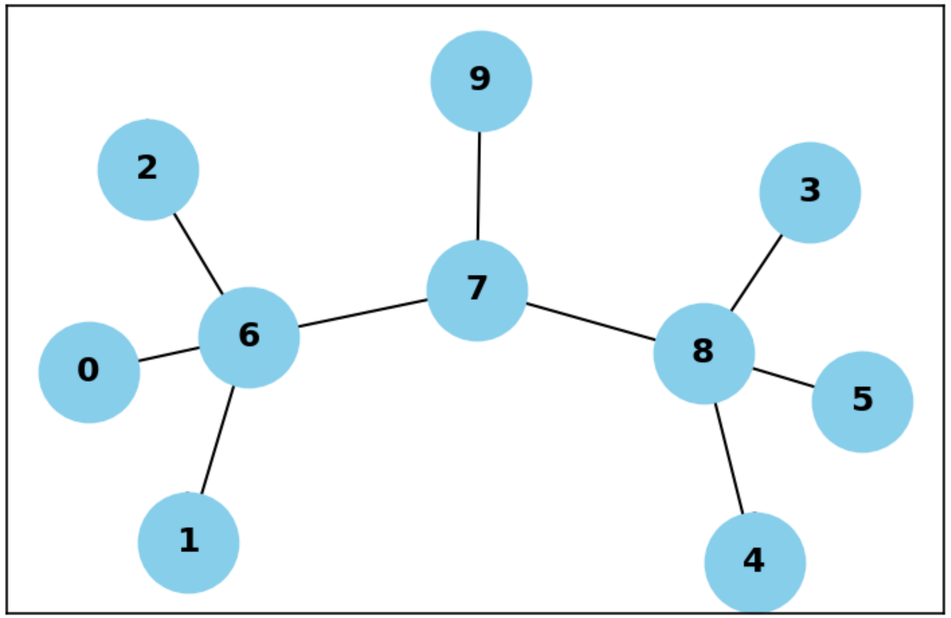


Graph neural networks basics recap (see Lecture 8)

Graph representation

- Nodes = Atoms
- Edges = Bonds
- G = (*E*, *V*) - graph
  - *V* - set of nodes v_i
  - *E* - set of edges $\{e_{ij}\}$ between the nodes

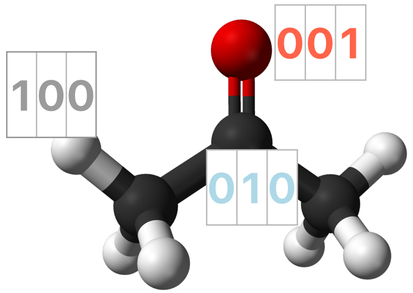


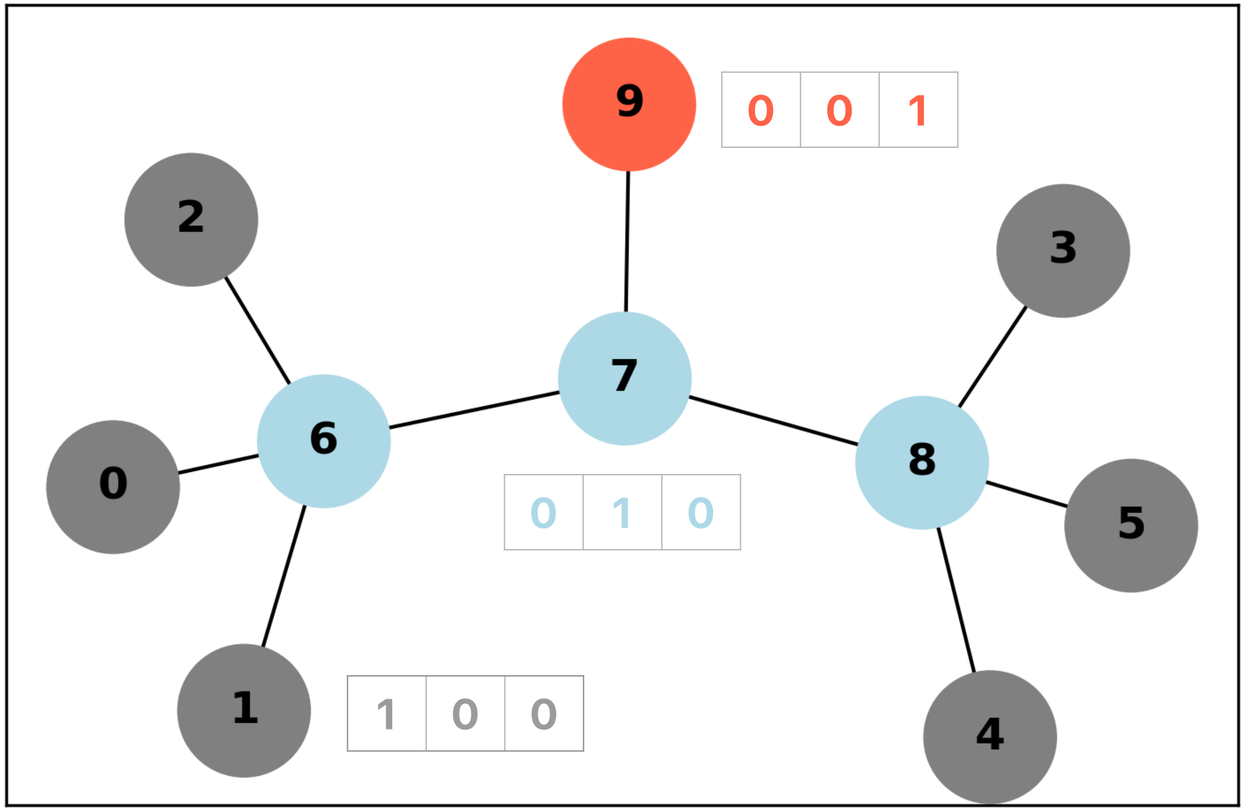


Hydrogen	1	0	0
Carbon	0	1	0
Oxygen	0	0	1

Node encoding

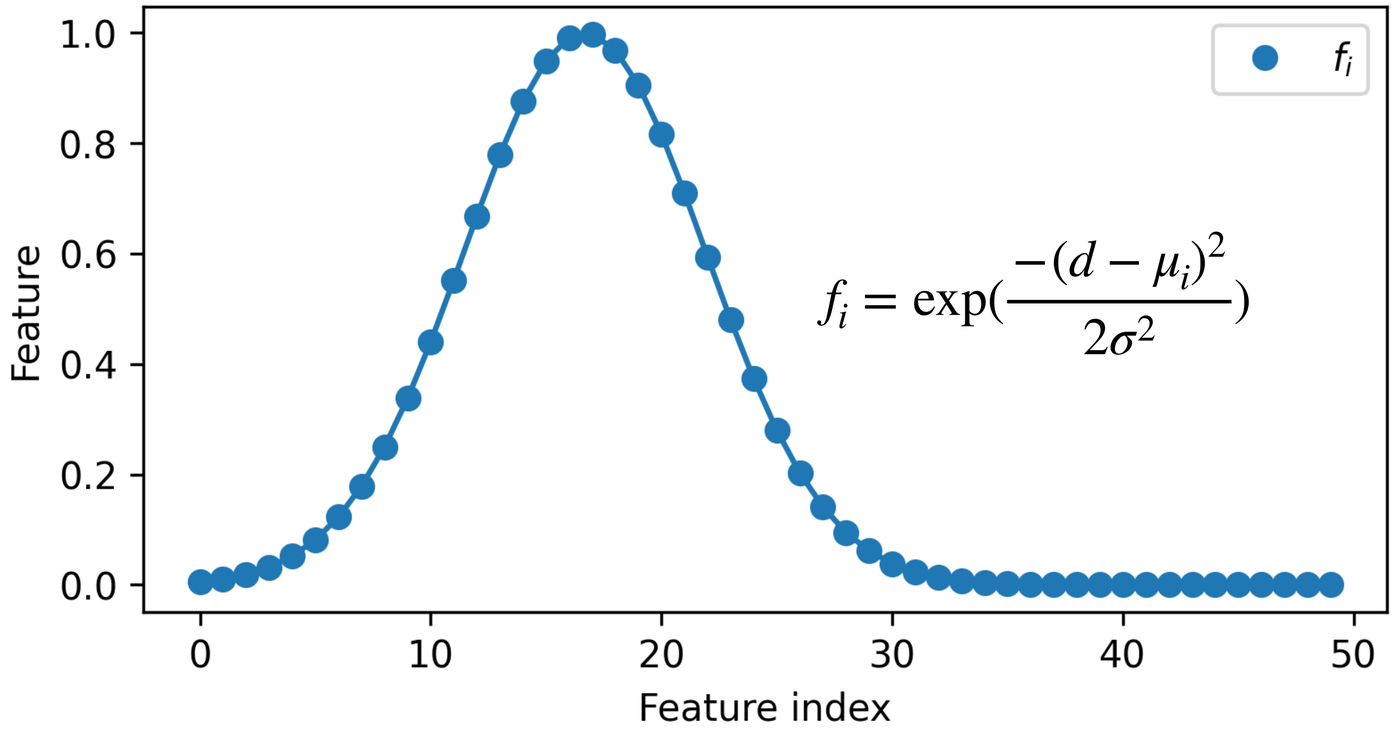
- One-Hot encoding





Edge encoding

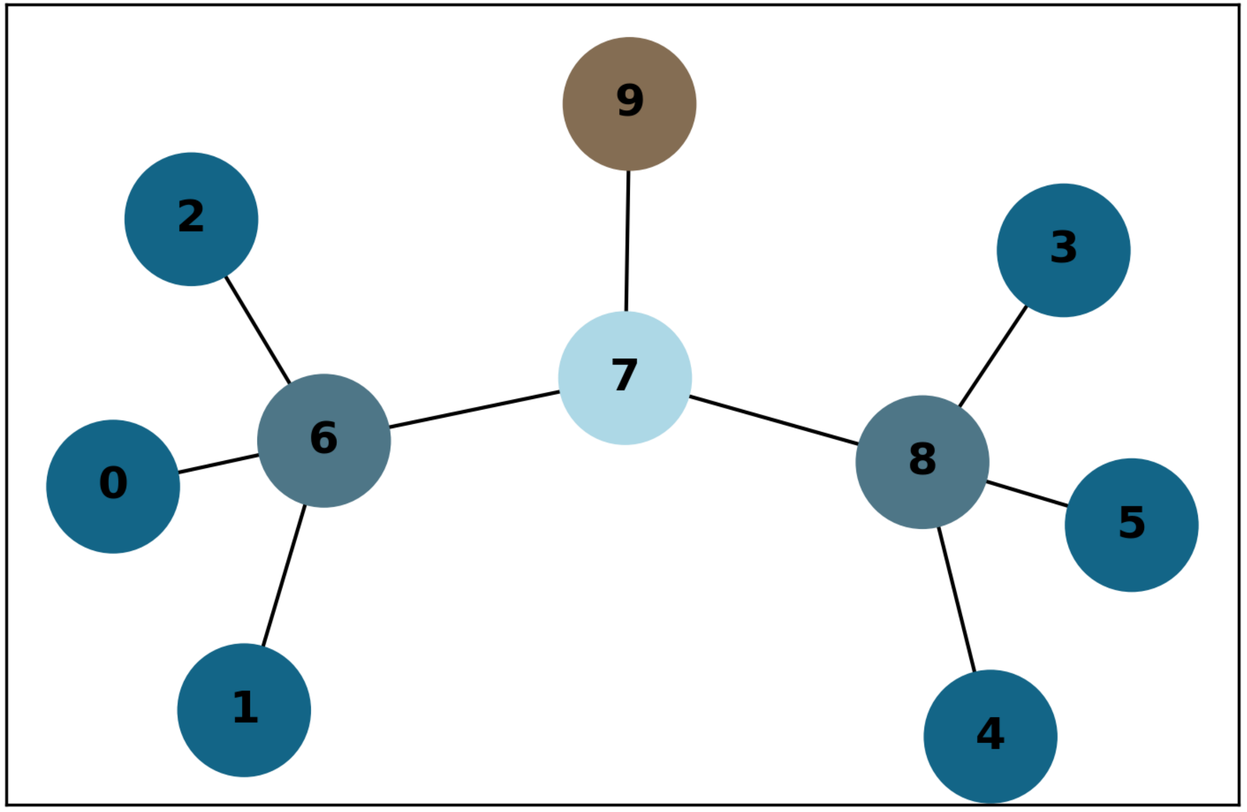
- Bond distance expansion
  - e.g. gaussian expansion
  - Edge e_{ij} feature = $[f_1^{ij}, \ldots, f_n^{ij}]$
  - Scalar features invariant to E(3)



Global pooling (readout) layer

- Pool updated node attributes into a single vector
- Using permutation invariant operation (e.g., min, max, mean, sum)

$$h = \frac{1}{N}\sum \mathbf{n_i} = \frac{1}{N} \cdot \begin{matrix} \mathbf{n^1_1} & \mathbf{n^1_2} & \mathbf{n^1_3} \\ + \\ \dots \\ + \\ \mathbf{n^9_1} & \mathbf{n^9_2} & \mathbf{n^9_3} \end{matrix}$$



Global pooling (readout) layer

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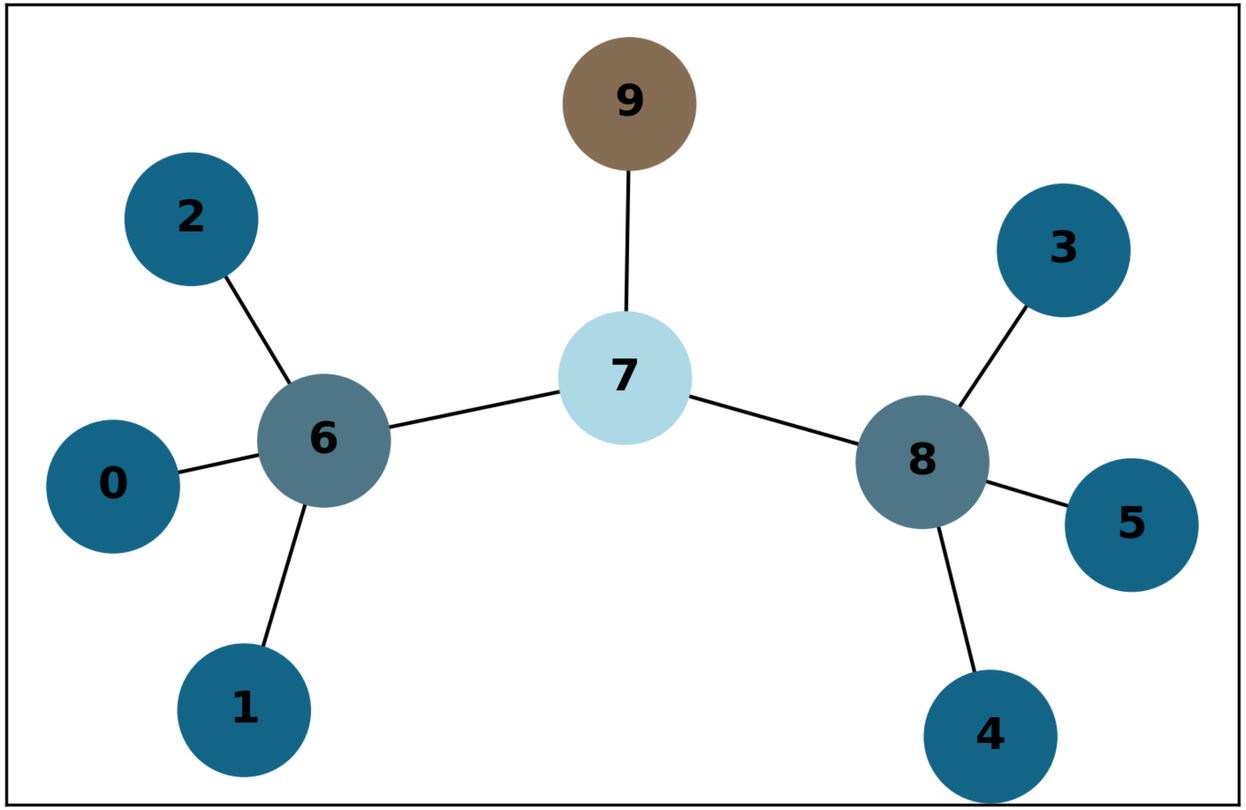
- Using permutation invariant operation (e.g., min, max, mean, sum)

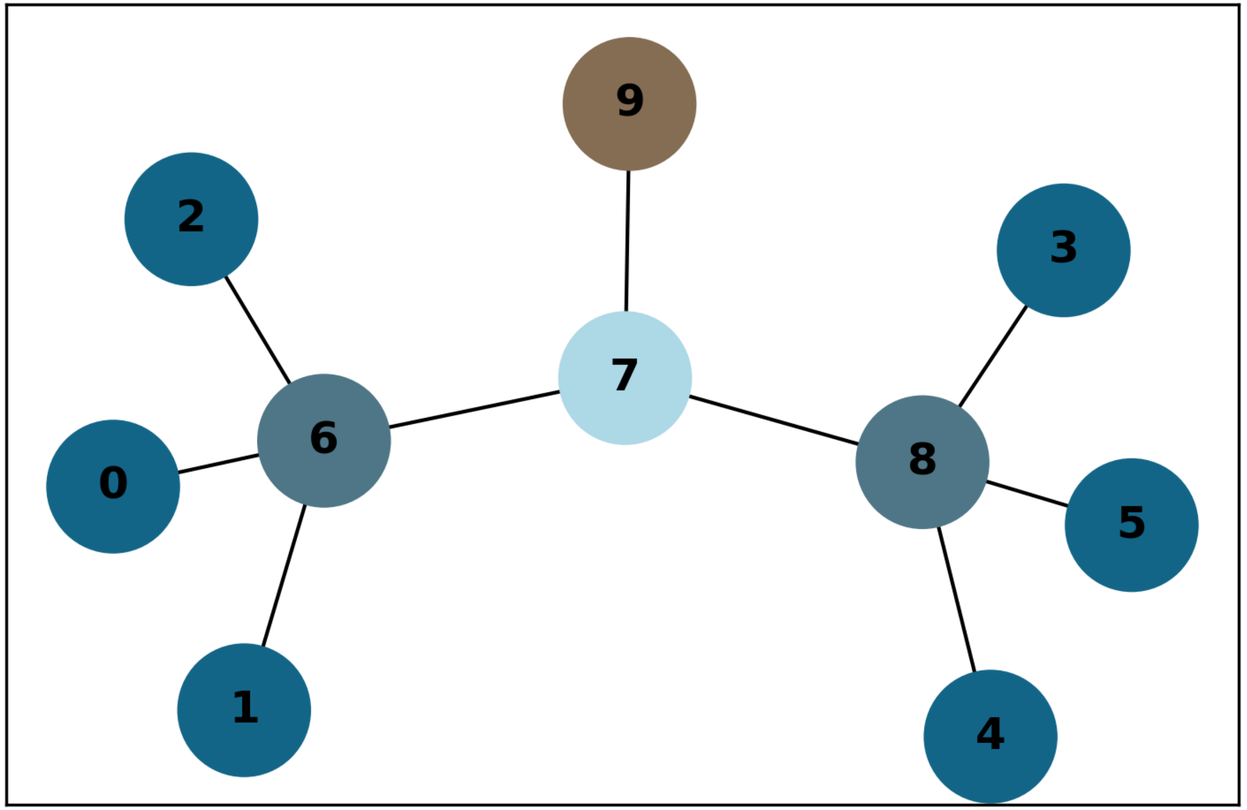
$$h = \frac{1}{N}\sum \mathbf{n_i} = \begin{array}{|c|c|c|}\hline h_1 & h_2 & h_3 \\ \hline \end{array}$$



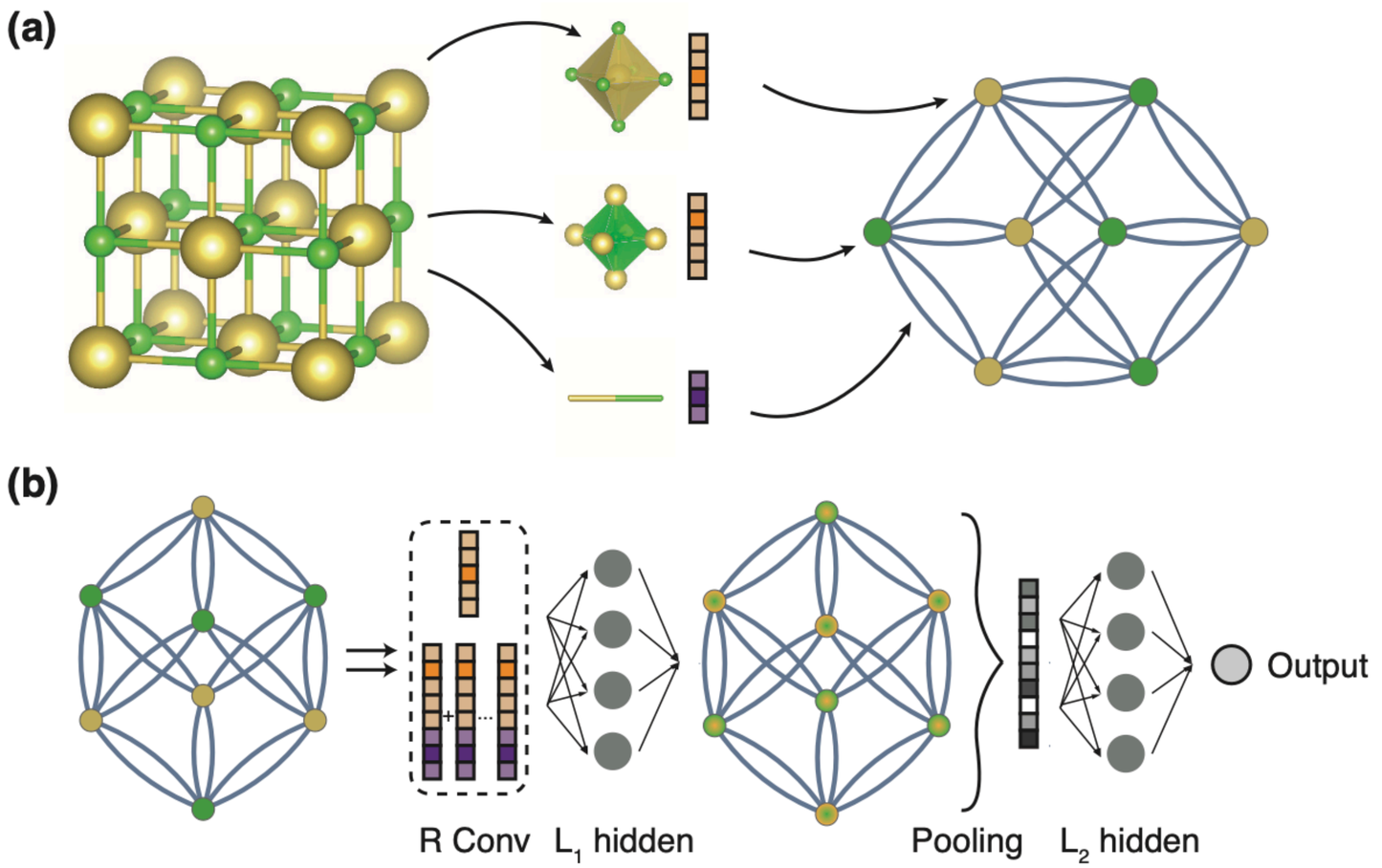
$$h = \frac{1}{N}\sum \mathbf{n_i} = \boxed{\mathbf{h_1}}\boxed{\mathbf{h_2}}\boxed{\mathbf{h_3}}$$

Final prediction

- MLP($\boxed{\mathbf{h_1}}\boxed{\mathbf{h_2}}\boxed{\mathbf{h_3}}$)
  - e.g., MLP(3 -> 16 -> 1)
  - 3 - input vector size
  - 16 - hidden layer size
  - 1 - output size



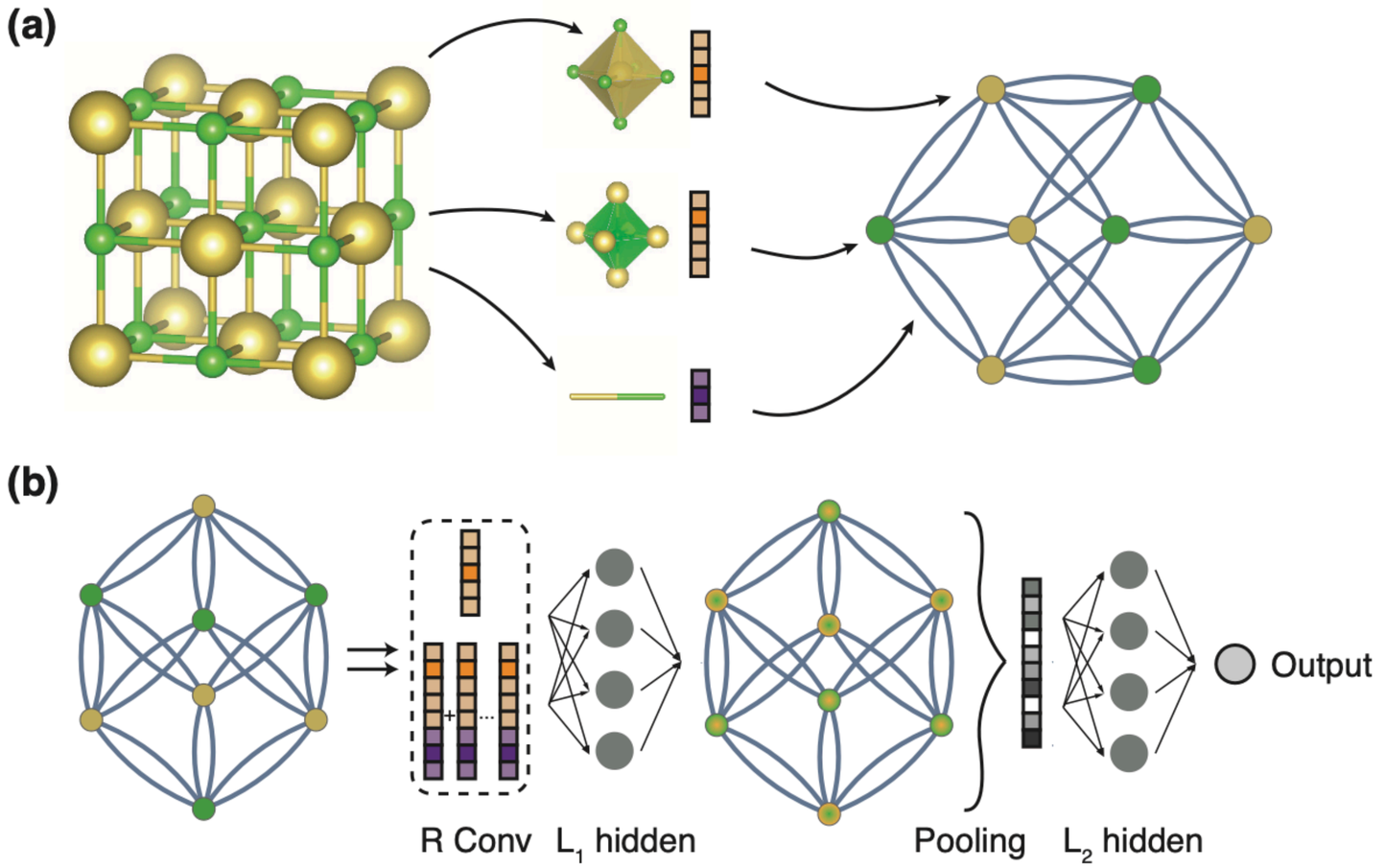
Example: Crystal graph convolutional neural network



Crystal Graph Convolutional Neural Networks for an Accurate and Interpretable Prediction of Material Properties

Pipeline

- Convert structure to graph
- Update node states (convolution)
- Apply pooling
- Apply post-processing layer (regression of classification)



Pipeline

- Convert structure to graph
- Update node states (convolution)
- Apply pooling
- Apply post-processing layer (regression of classification)

$$\boldsymbol{v}_i^{(t+1)} = \text{Conv}\left(\boldsymbol{v}_i^{(t)}, \boldsymbol{v}_j^{(t)}, \boldsymbol{u}_{(i,j)_k}\right), (i,j)_k \in \mathcal{G}$$

$$\boldsymbol{v_c} = \text{Pool}\left(\boldsymbol{v}_0^{(0)}, \boldsymbol{v}_1^{(0)}, ..., \boldsymbol{v}_N^{(0)}, ..., \boldsymbol{v}_N^{(R)}\right)$$

Pipeline

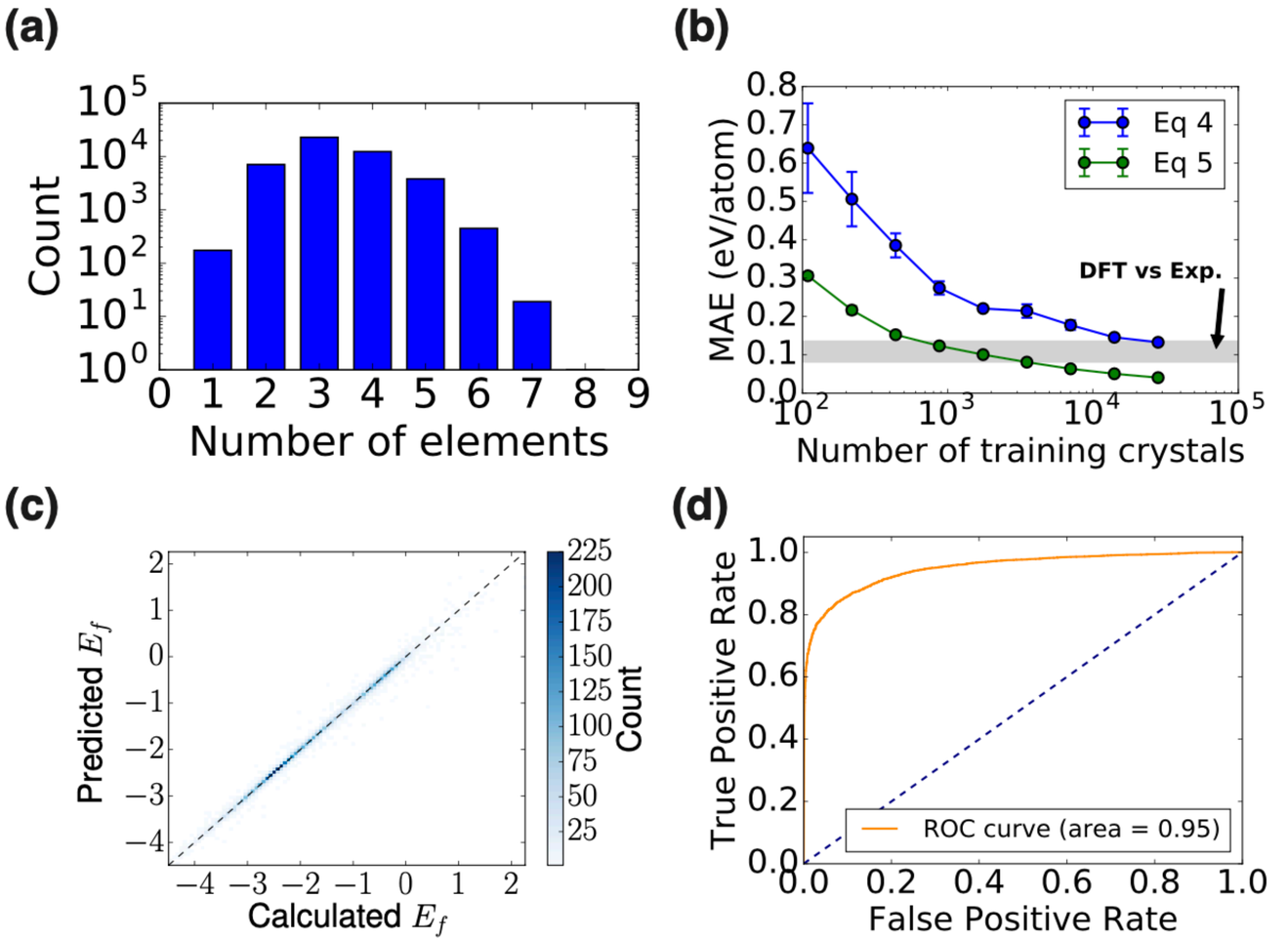
- Convert structure to graph
- Update node states
- Apply pooling
- Apply post-processing layer (regression of classification)

$$\boldsymbol{z}_{(i,j)_k}^{(t)} = \boldsymbol{v}_i^{(t)} \oplus \boldsymbol{v}_j^{(t)} \oplus \boldsymbol{u}_{(i,j)_k}$$

$$\boldsymbol{v}_i^{(t+1)} = \boldsymbol{v}_i^{(t)} + \sum_{j,k} \sigma(\boldsymbol{z}_{(i,j)_k}^{(t)} \boldsymbol{W}_f^{(t)} + \boldsymbol{b}_f^{(t)}) \odot g(\boldsymbol{z}_{(i,j)_k}^{(t)} \boldsymbol{W}_s^{(t)} + \boldsymbol{b}_s^{(t)})$$

$$\boldsymbol{v}_c = \text{Pool}\left(\boldsymbol{v}_0^{(0)}, \boldsymbol{v}_1^{(0)}, ..., \boldsymbol{v}_N^{(0)}, ..., \boldsymbol{v}_N^{(R)}\right)$$

Property	# of train data	Unit	MAE_{model}	MAE_{DFT}
Formation energy	28046	eV/atom	0.039	0.081–0.136[18]
Absolute energy	28046	eV/atom	0.072	–
Band gap	16458	eV	0.388	0.6[23]
Fermi energy	28046	eV	0.363	–
Bulk moduli	2041	log(GPa)	0.054	0.050[24]
Shear moduli	2041	log(GPa)	0.087	0.069[24]
Poisson ratio	2041	–	0.030	–



Equivariance of force prediction

$$E = \sum_{i \in N_{atoms}} E_{i,atomic}$$

$$\overrightarrow{F}_i = -\nabla_i E$$

Equivariance of force prediction

$$r_{ij} = |\mathbf{r}_i - \mathbf{r}_j|$$

$$\phi_k(r_{ij}) = \exp\left[-\frac{(r_{ij} - \mu_k)^2}{2\sigma_k^2}\right]$$

$$E = \sum_{i \in N_{atoms}} E_{i,atomic}$$

$$\overrightarrow{F}_i = -\nabla_i E$$

Equivariance of force prediction

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$$r_{ij} = |\mathbf{r}_i - \mathbf{r}_j| \qquad \phi_k(r_{ij}) = \exp\left[-\frac{(r_{ij} - \mu_k)^2}{2\sigma_k^2}\right]$$

$$\mathbf{h}_i^{(t+1)} = \text{Update}\left(\mathbf{h}_i^{(t)}, \sum_{j \in \mathcal{N}(i)} \text{Message}\left(\mathbf{h}_i^{(t)}, \mathbf{h}_j^{(t)}, \boldsymbol{\phi}(r_{ij})\right)\right)$$

Equivariance of force prediction

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$$E = \sum_i \varepsilon(\mathbf{h}_i^{(T)})$$

$$\frac{\partial E}{\partial \mathbf{r}_i} = \sum_{j \neq i} \frac{\partial E}{\partial r_{ij}} \frac{\mathbf{r}_i - \mathbf{r}_j}{r_{ij}}$$

Equivariance of force prediction

$$\frac{\partial E}{\partial \mathbf{r}_i} = \sum_{j \neq i} \frac{\partial E}{\partial r_{ij}} \frac{\mathbf{r}_i - \mathbf{r}_j}{r_{ij}}$$

$$E = \sum_{i \in N_{atoms}} E_{i,atomic}$$

$$E(Q\mathbf{r} + \mathbf{t}) = E(\mathbf{r}) \qquad \mathbf{F}_i(Q\mathbf{r} + \mathbf{t}) = Q\mathbf{F}_i(\mathbf{r})$$

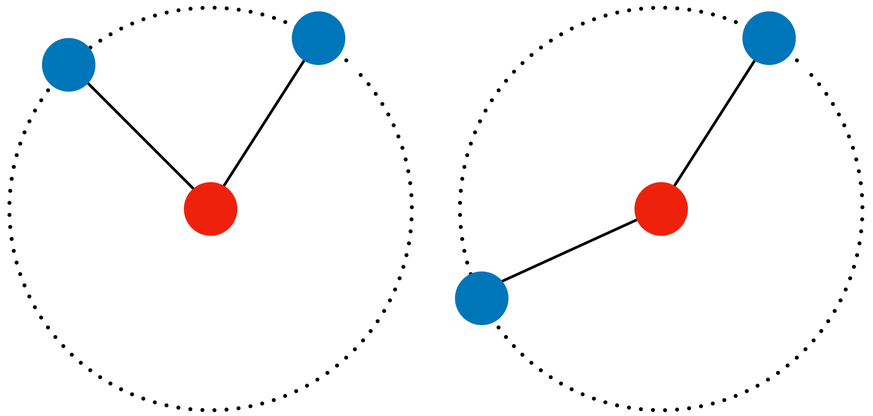
$$\overrightarrow{F}_i = -\nabla_i E$$

“Simple“ GNNs limitations

- Expressivity limit - different structures may have the same graph
  - E.g., graphs don’t know about angles

- Indistinguishable geometries for the discussed graph representation

- The same energies will be predicted for this molecule

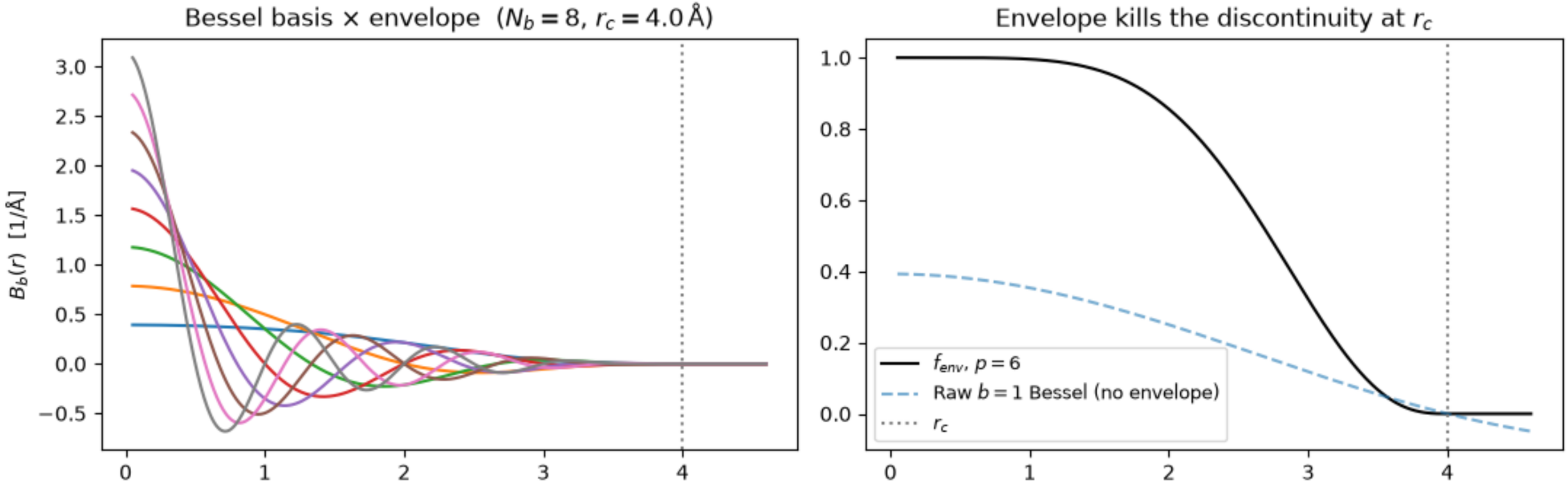


Concept behind equivariant GNNs

Distance encoding

- Radial Bessel basis
- Invariant to translation

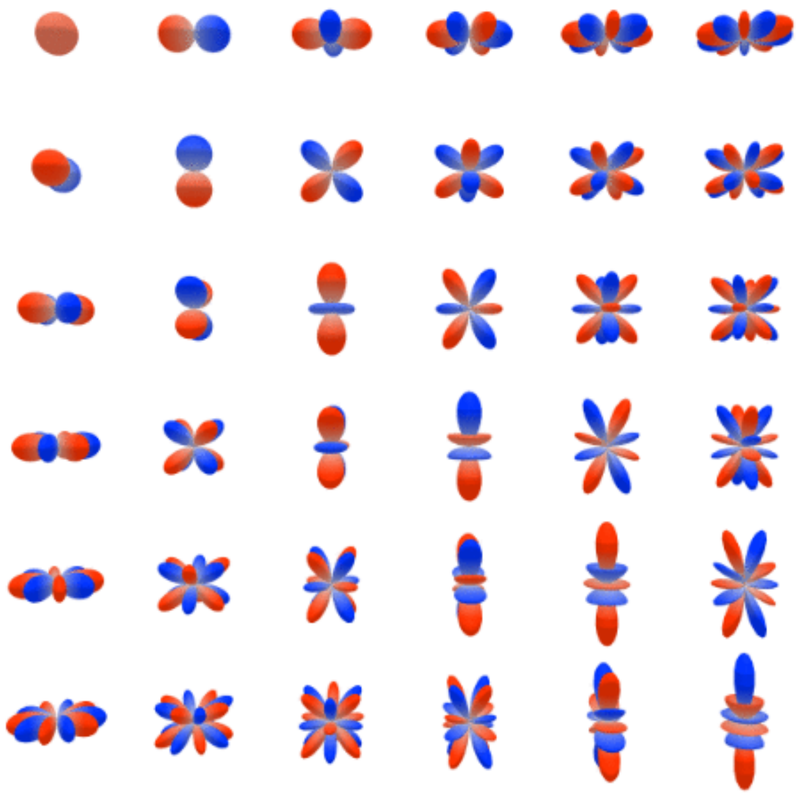
$$B_n(r_{ij}) = \frac{2}{r_c} \frac{\sin(\frac{n\pi}{r_c} r_{ij})}{r_{ij}} f_{env}(r_{ij}, r_c)$$



Direction encoding

- Spherical harmonics
- Rotation equivariant

$$Y_m^{(l)}(\hat{r}_{ij})$$



Convolution filter

- Spherical harmonics
- Rotation equivariant

$$S_m^{(l)}(\vec{r}_{ij}) = R(r_{ij})Y_m^{(l)}(\hat{r}_{ij})$$

$$R(r_{ij}) = W_n\sigma(\ldots\sigma(W_2\sigma(W_1B(r_{ij}))))$$

Node update

- Clebsch–Gordan coefficients are the “glue" that guarantees the message transforms correctly (equivariantly)

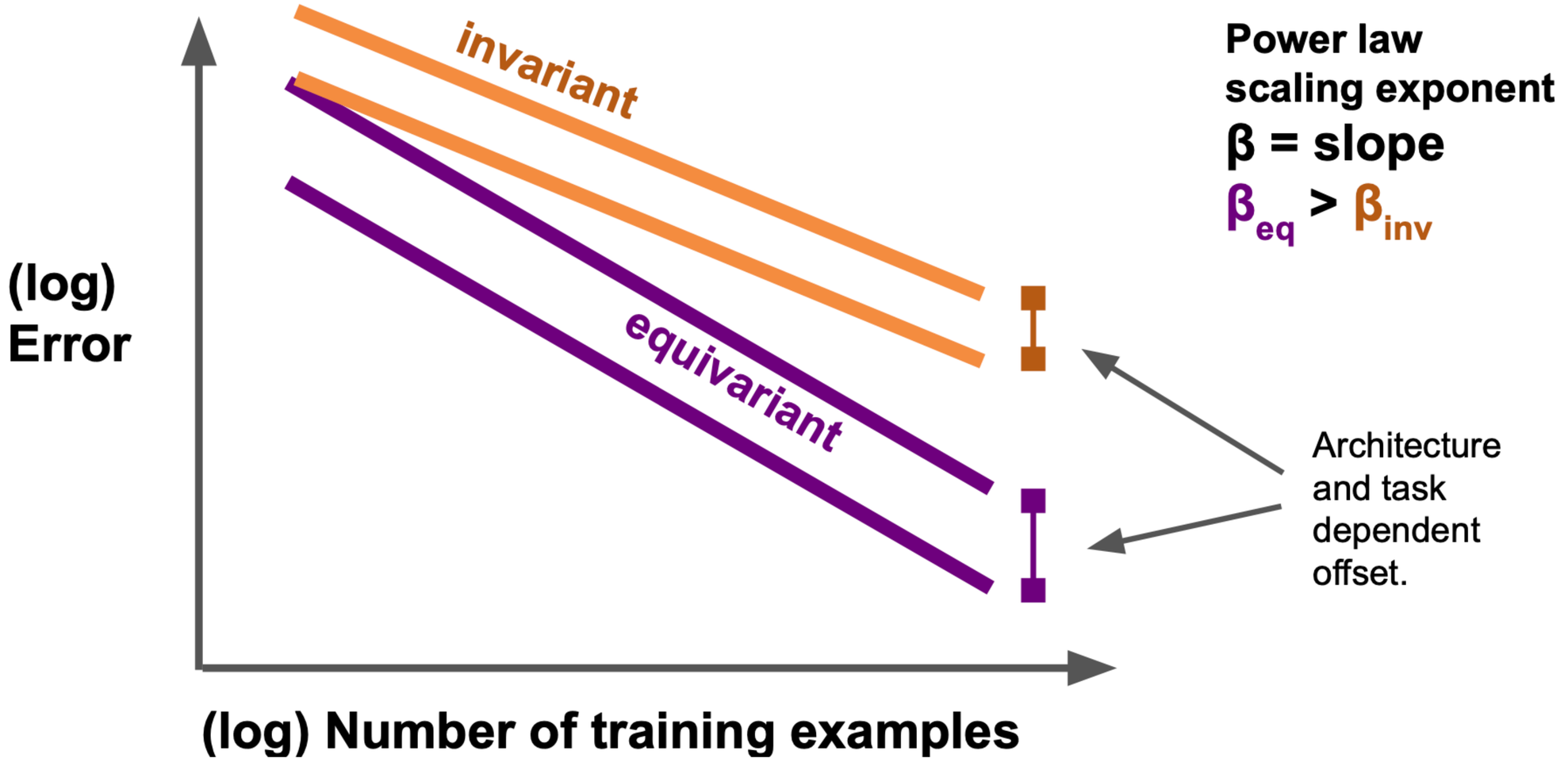
Clebsch–Gordan coefficients

Node features

$$m_{ij,m_o}^{(l_o)} = \sum_{l_f,l} \sum_{m_f,m} C_{(l,m)(l_f,m_f)}^{(l_o,m_o)} \, S_m^{(l)}(\vec{r}_{ij}) \, W^{(l,l_f)} \, f_{j,m_f}^{(l_f)},$$

$$M_{i,m_o}^{(l_o)} = \sum_{j \in \mathcal{N}(i)} m_{ij,m_o}^{(l_o)},$$

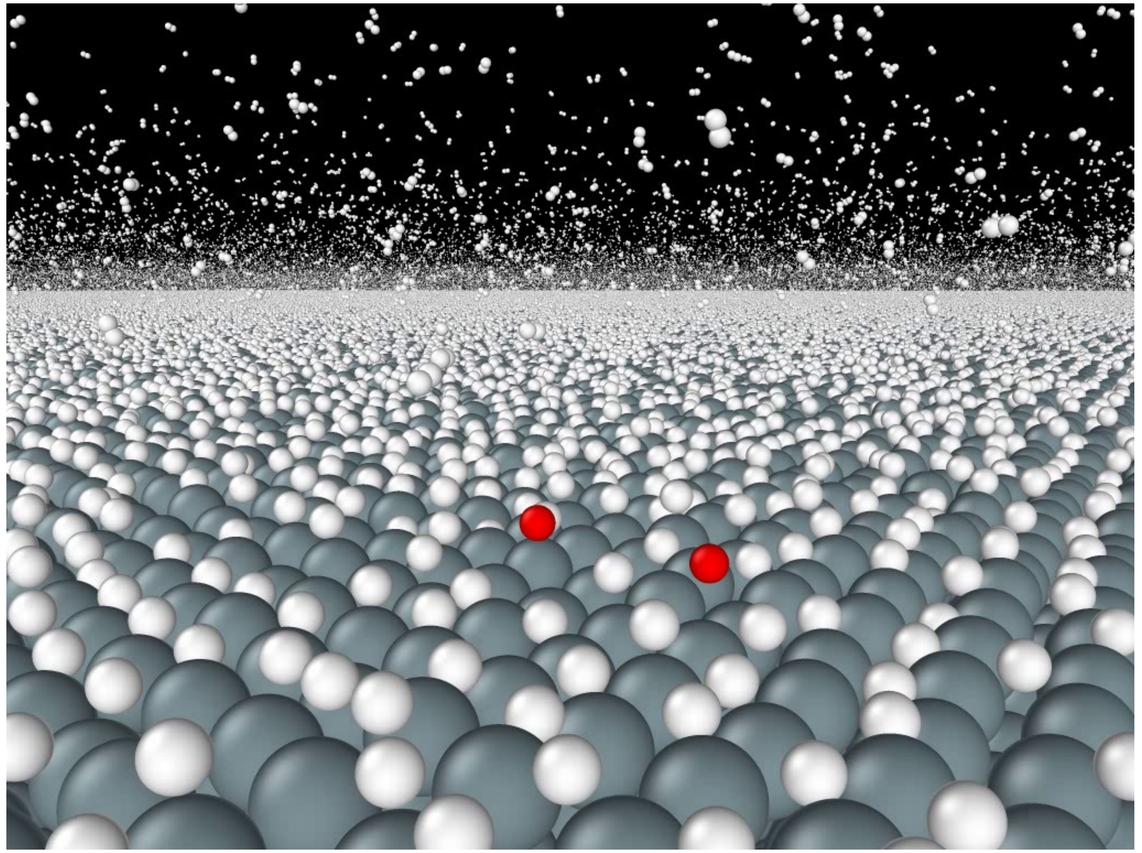
$$f_{i,m_o}^{(l_o)} = \sigma\left(\sum_{l_f} W_{\text{self}}^{(l_o,l_f)} \, f_{i,m_o}^{(l_f)} + M_{i,m_o}^{(l_o)} \right).$$


invariant
equivariant
(log)
Error
(log) Number of training examples
Power law
scaling exponent
β = slope
$\beta_{eq} > \beta_{inv}$
Architecture
and task
dependent
offset.

MLIPs enable LARGE-SCALE simulations

- 0.5 trillion atoms on 27,336 NVIDIA V100 GPUs

- Just unthinkable for direct DFT simulations



Source: https://cfp.scipy.org/2023/talk/BDV3EE/

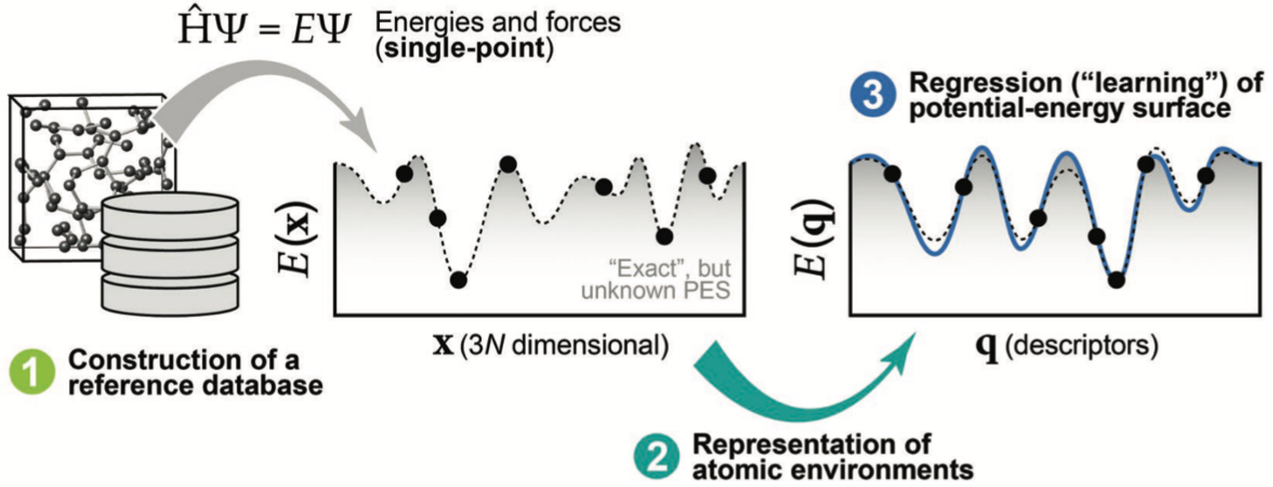
Fitting (training) interatomic potentials

Fitting (training) interatomic potentials

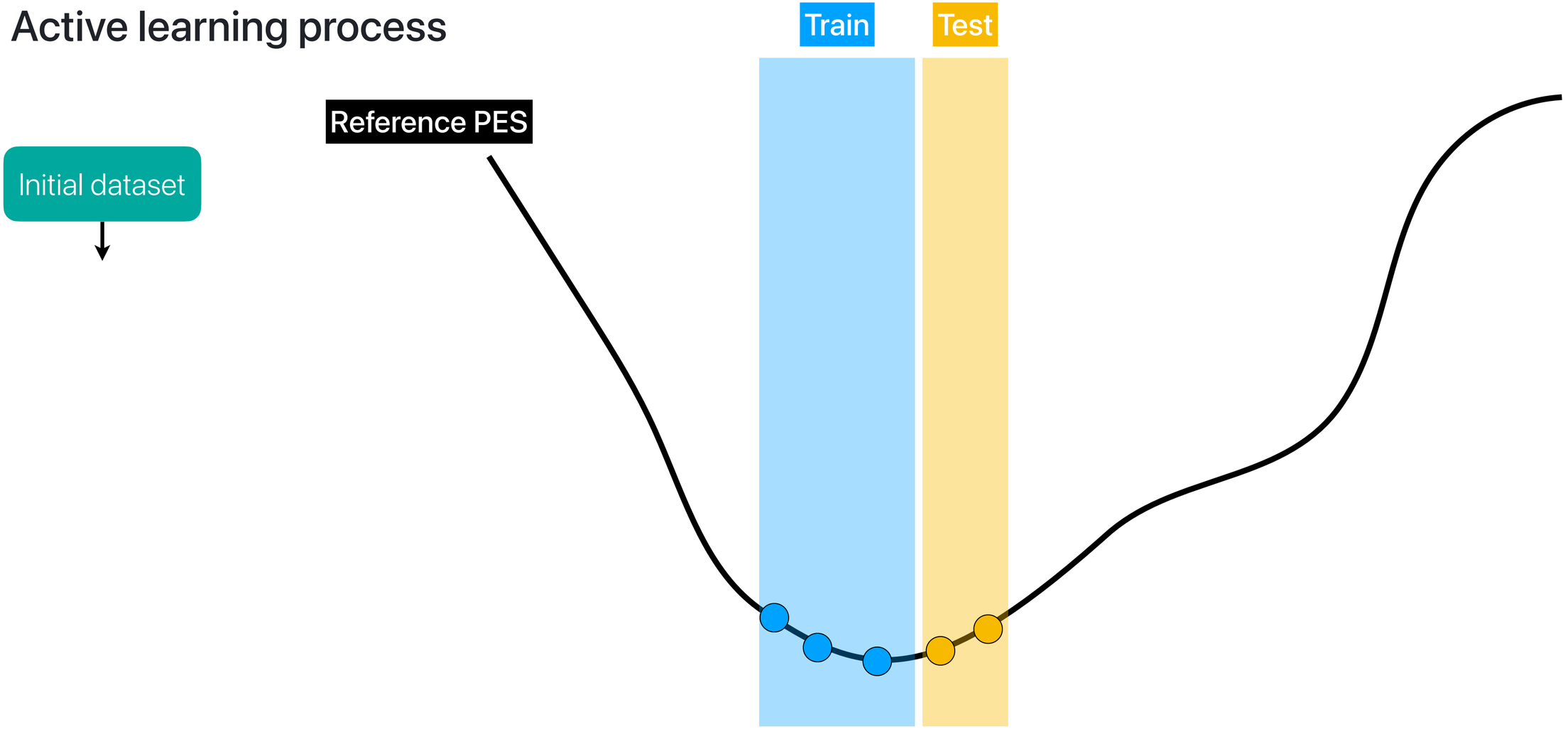
- For a **given system**
  - We run AIMD at elevated temperatures to sample atomic configurations
- The dataset
  - Geometries: $\{R_n\}$
  - Energies: $\{E_n(R_n)\}$
  - Forces: $\{F_{i,\alpha}(R_n)\}$
- Loss function

$$\mathscr{L} = \lambda_E \left|\left| \hat{E} - E \right|\right|^2 + \lambda_F \frac{1}{3N} \sum_{i=1}^{N} \sum_{\alpha=1}^{3} \left|\left| -\frac{\partial \hat{E}}{\partial r_{i,\alpha}} - F_{i,\alpha} \right|\right|^2$$

  - The hat symbol denotes predictions

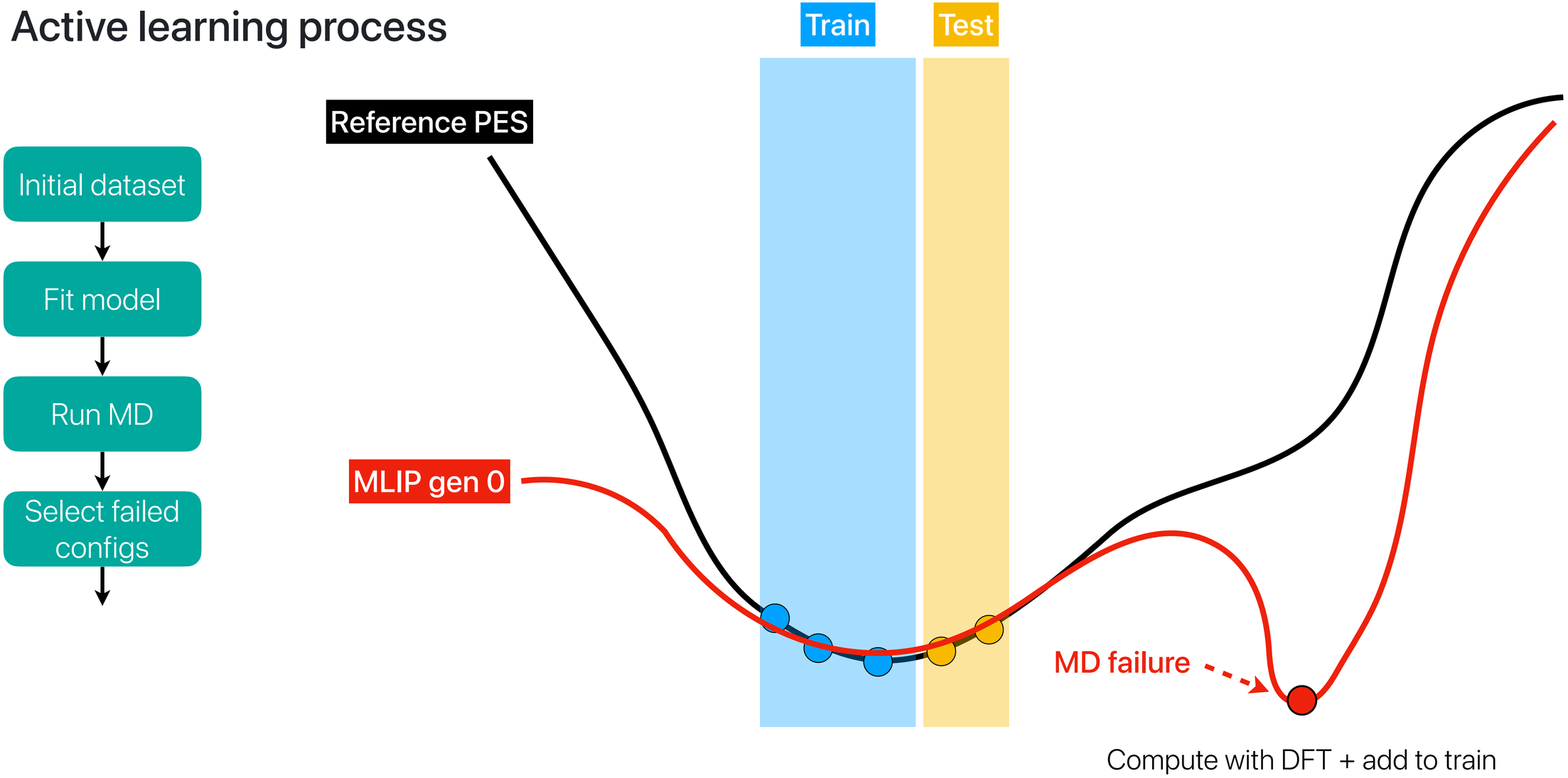


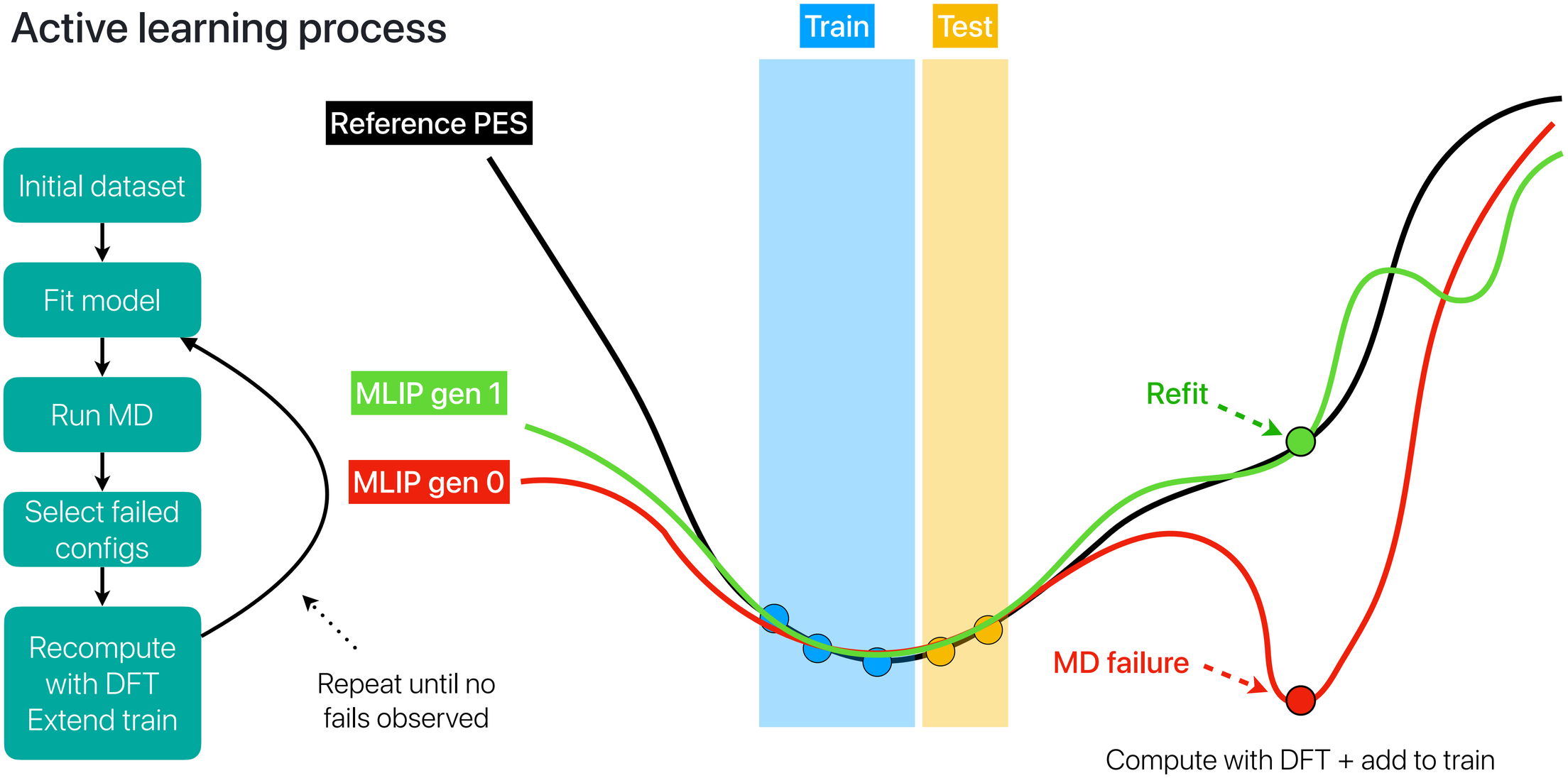
Active learning process



Reproduced from Tutorial 2: MACE active learning and fine-tuning: https://mace-docs.readthedocs.io/en/latest/examples/tutorials.html

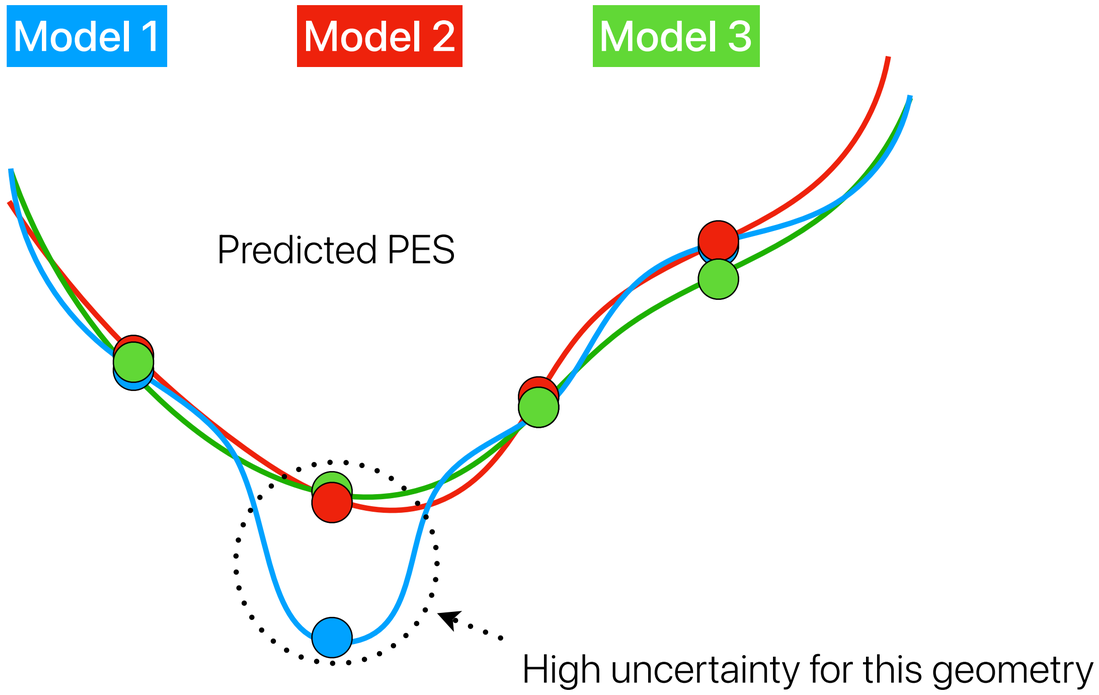
Active learning process




Active learning process
Initial dataset
Fit model
Run MD
Select failed configs
Recompute with DFT Extend train
Repeat until no fails observed
Reference PES
MLIP gen 1
MLIP gen 0
Train
Test
Refit
MD failure
Compute with DFT + add to train

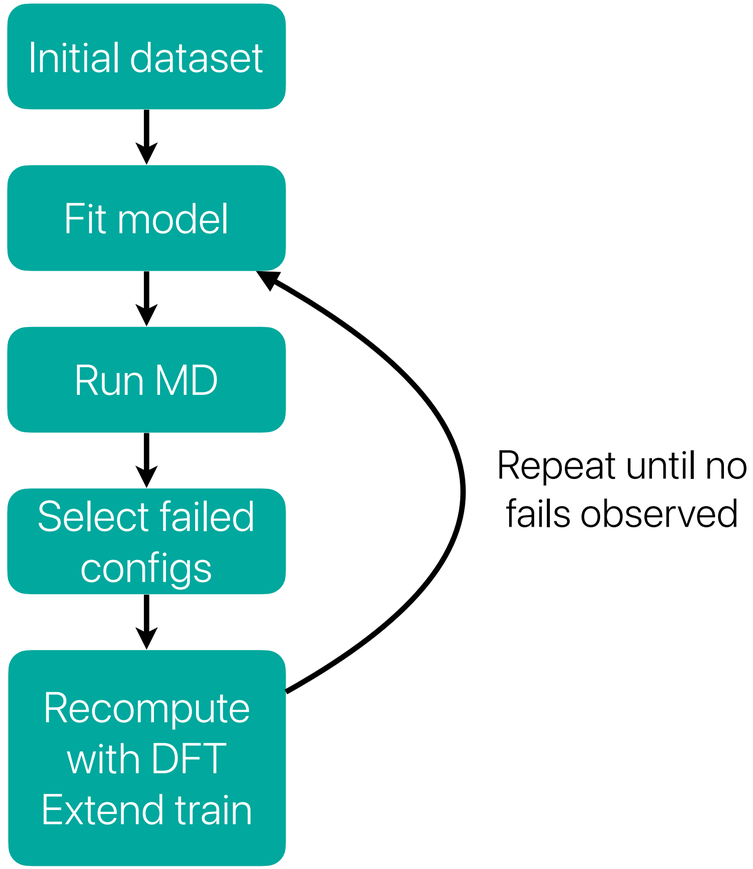
How MLIP failure during MD is detected?

- Committee disagreement
  - Ensemble of models -> std in energy/ forces prediction serves as uncertainty
- Or MaxVol/D-optimality criterion
  - Extrapolation grade (γ) from active set → $\gamma > tr$ signals extrapolation

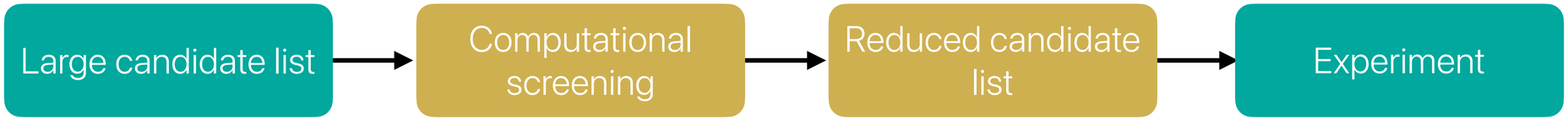


All in all

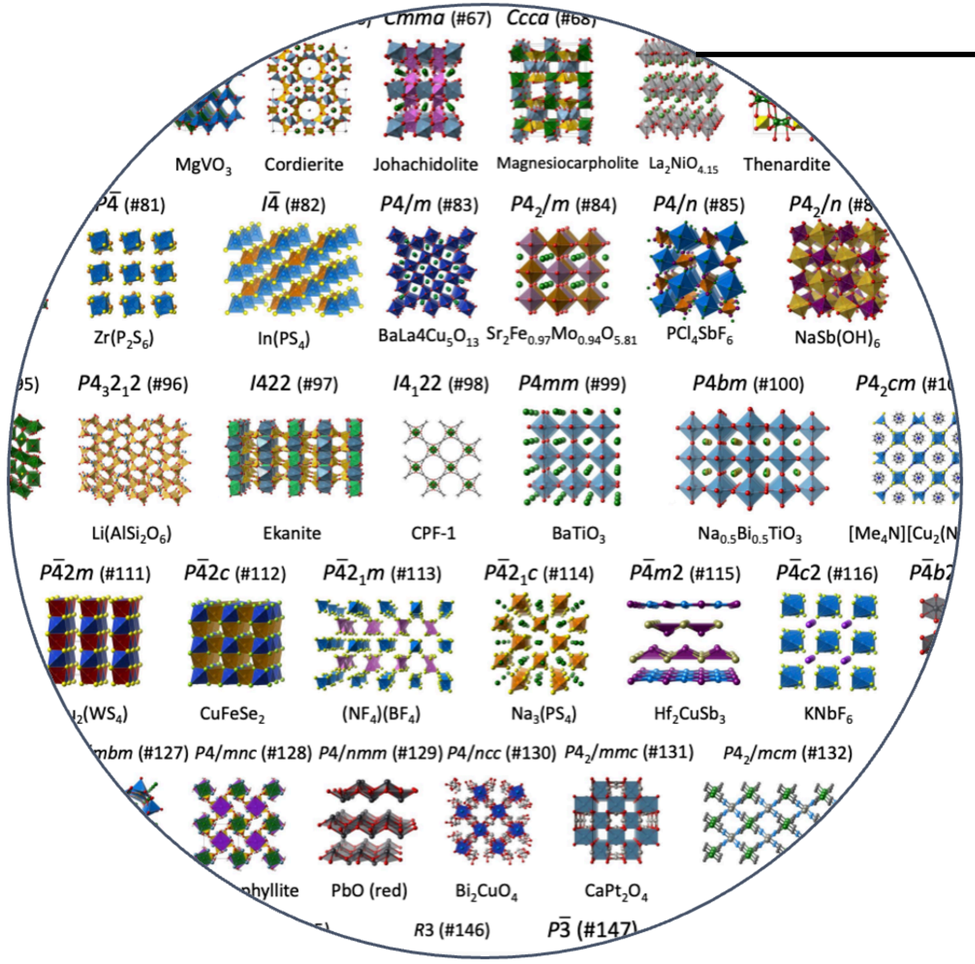
- Developing system-specific MLIP requires
  - Generating initial reference data
  - Fitting a model
  - Running MD for active learning stage
  - Once required accuracy is achieved the model can be used for the inference
- Takes some time (~1-3 weeks)



Funnel approach for materials selection



- We want to scan a large chemical space
- It is very inefficient to train system-specific MLIP for each candidate



Computational screening → Reduced candidate list →

We want a single model for:

- accurate description of many materials and chemical systems
- which generalizes to unseen structures, defects, etc

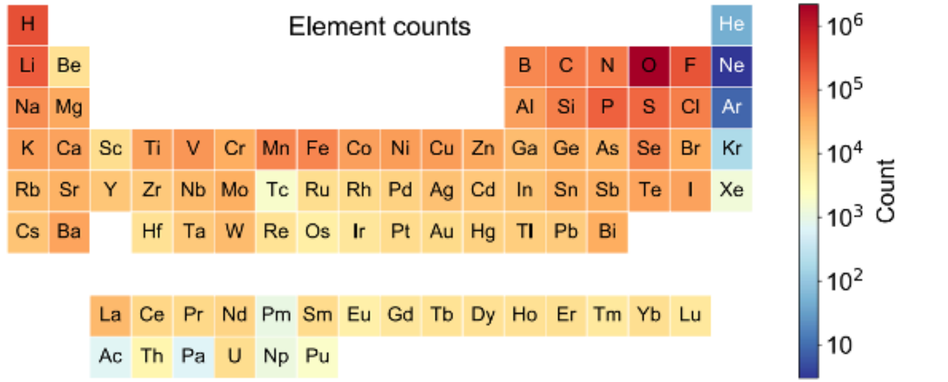
$$F(\{Z, \mathbf{R}\}, W) \xrightarrow{\mathbf{W}} E^*$$

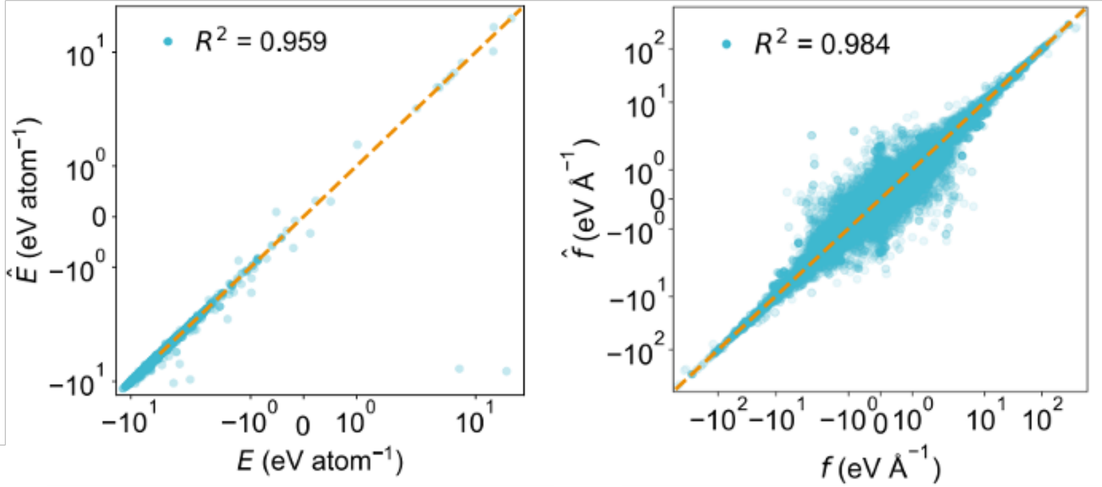
Article | Published: 28 November 2022

A universal graph deep learning interatomic potential for the periodic table

Chi Chen ✉ & Shyue Ping Ong ✉

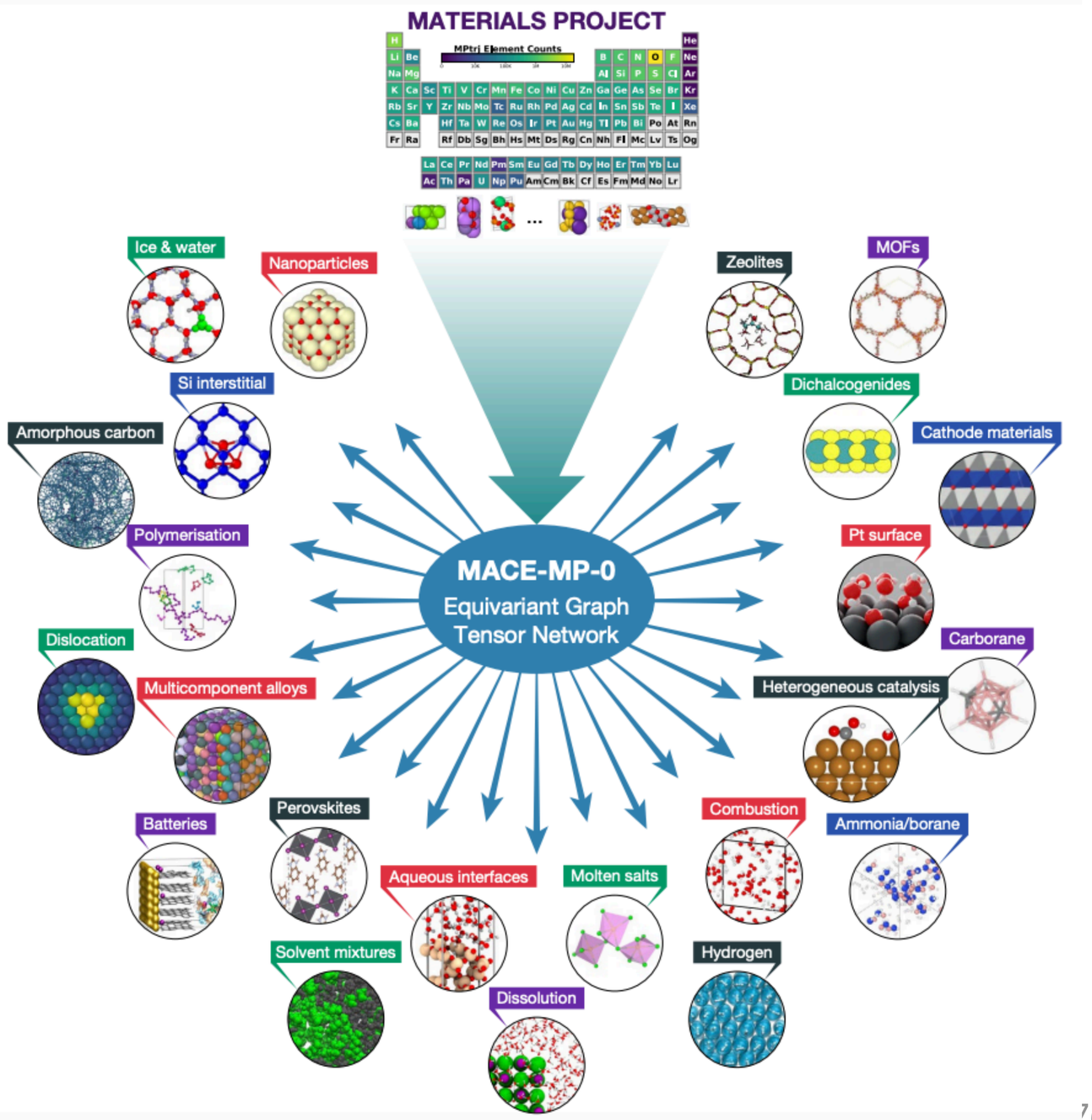
Nature Computational Science **2**, 718–728 (2022) | Cite this article

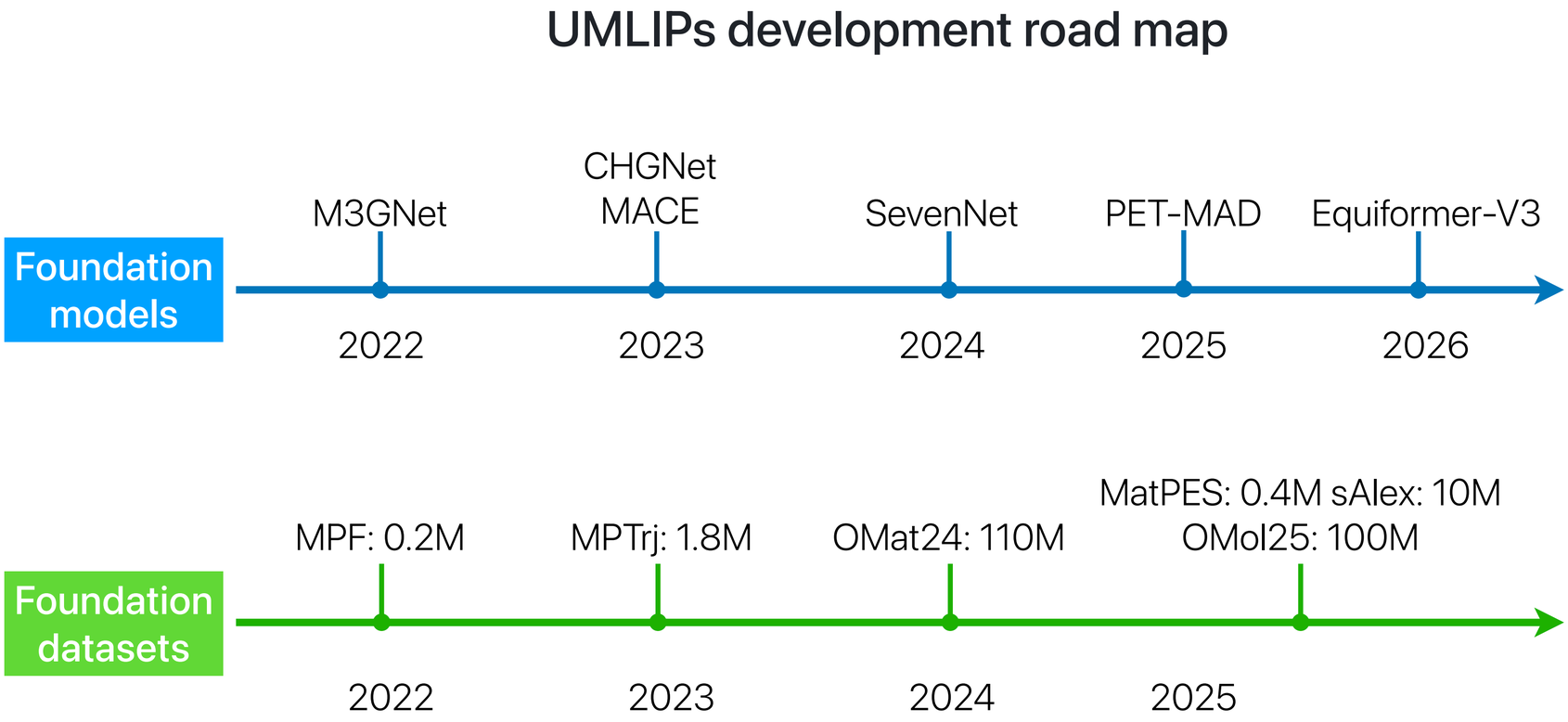




Supposed to be accurate for various downstream task

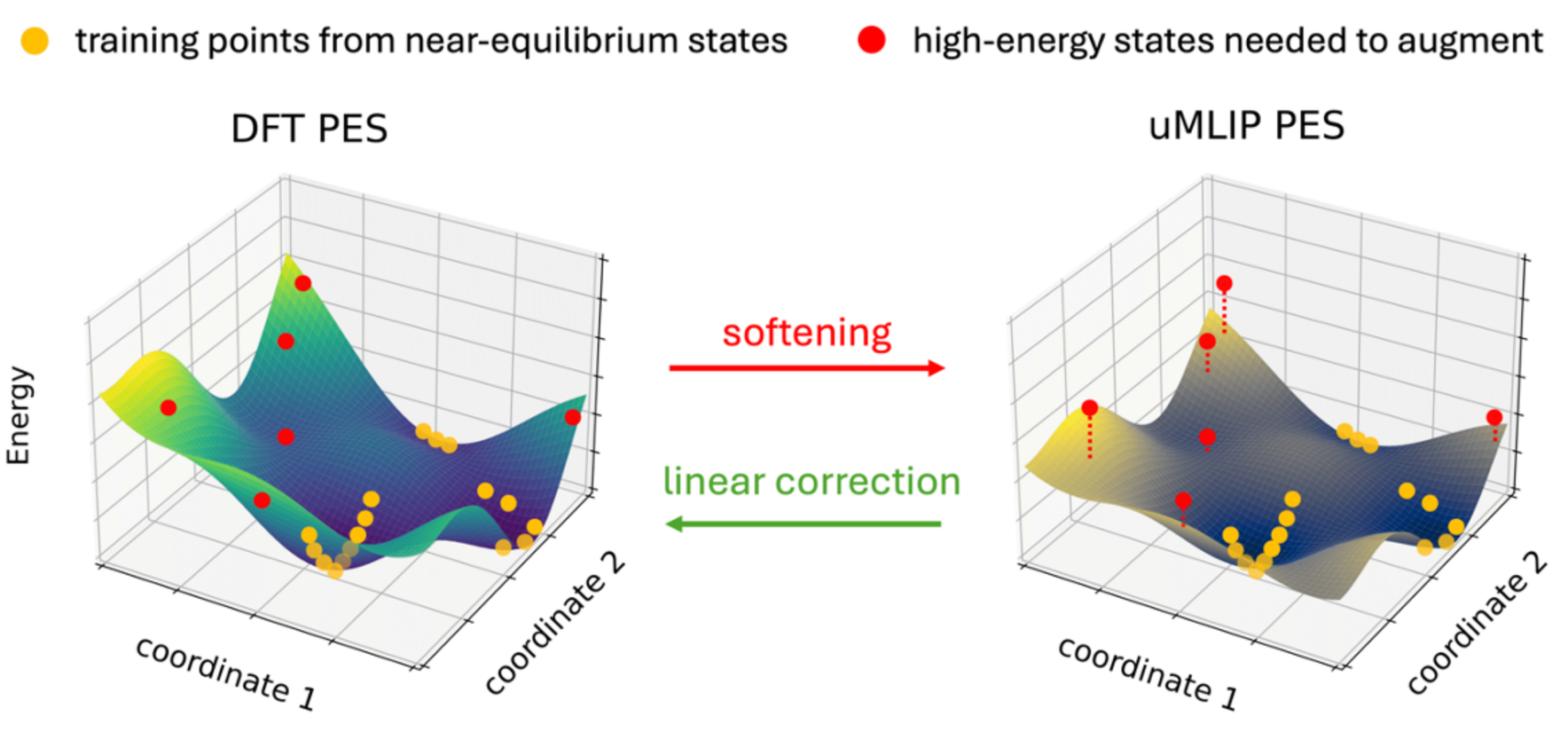
- Diffusion
- Adsorption
- Dissolution
- ...




UMLIPs development road map
Foundation models
M3GNet
CHGNet
MACE
SevenNet
PET-MAD
Equiformer-V3
2022
2023
2024
2025
2026
Foundation datasets
MPF: 0.2M
MPTrj: 1.8M
OMat24: 110M
MatPES: 0.4M sAlex: 10M
OMol25: 100M
2022
2023
2024
2025

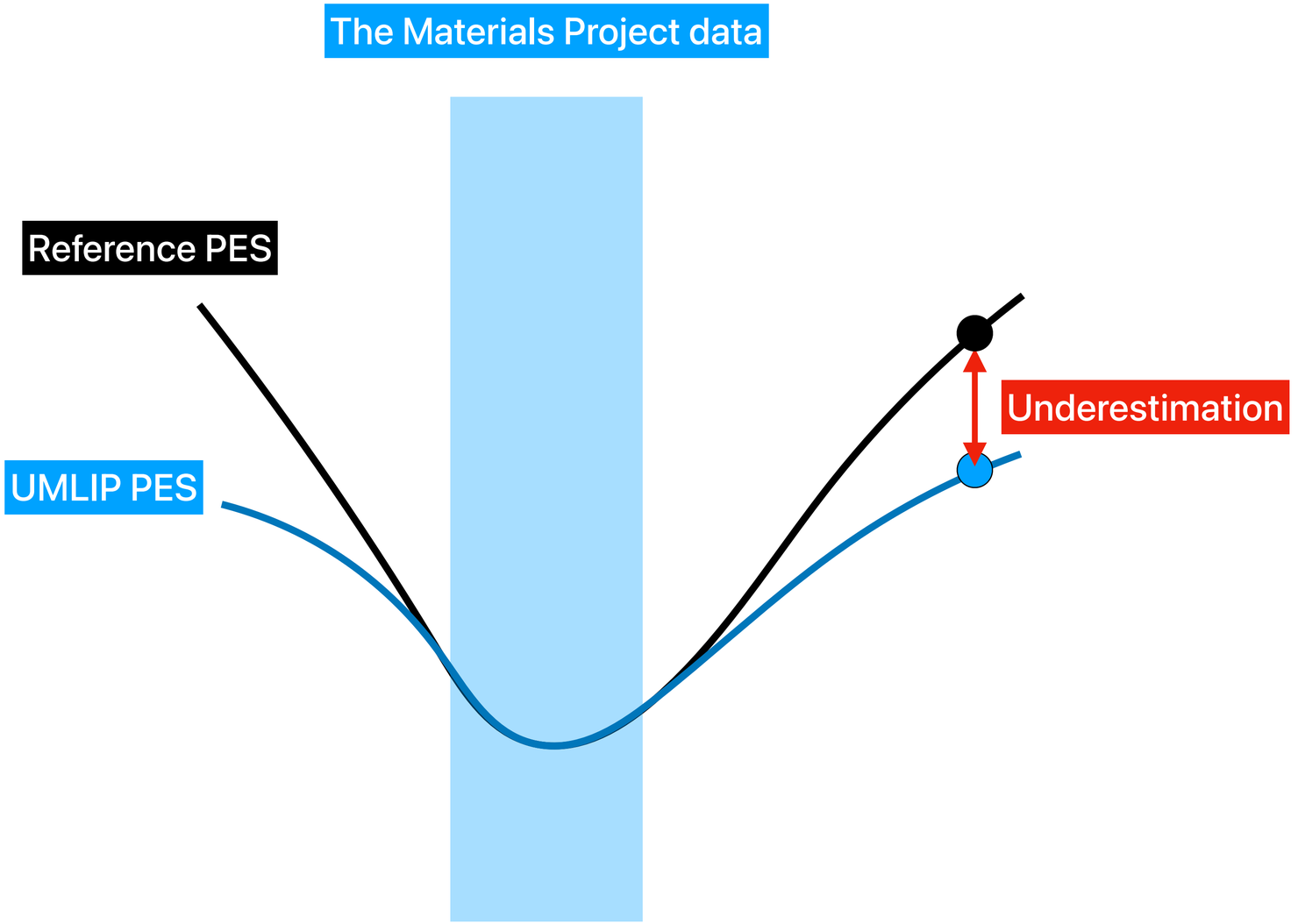
Systematic softening

- Near equilibrium training data
- Under-prediction for high-energy states



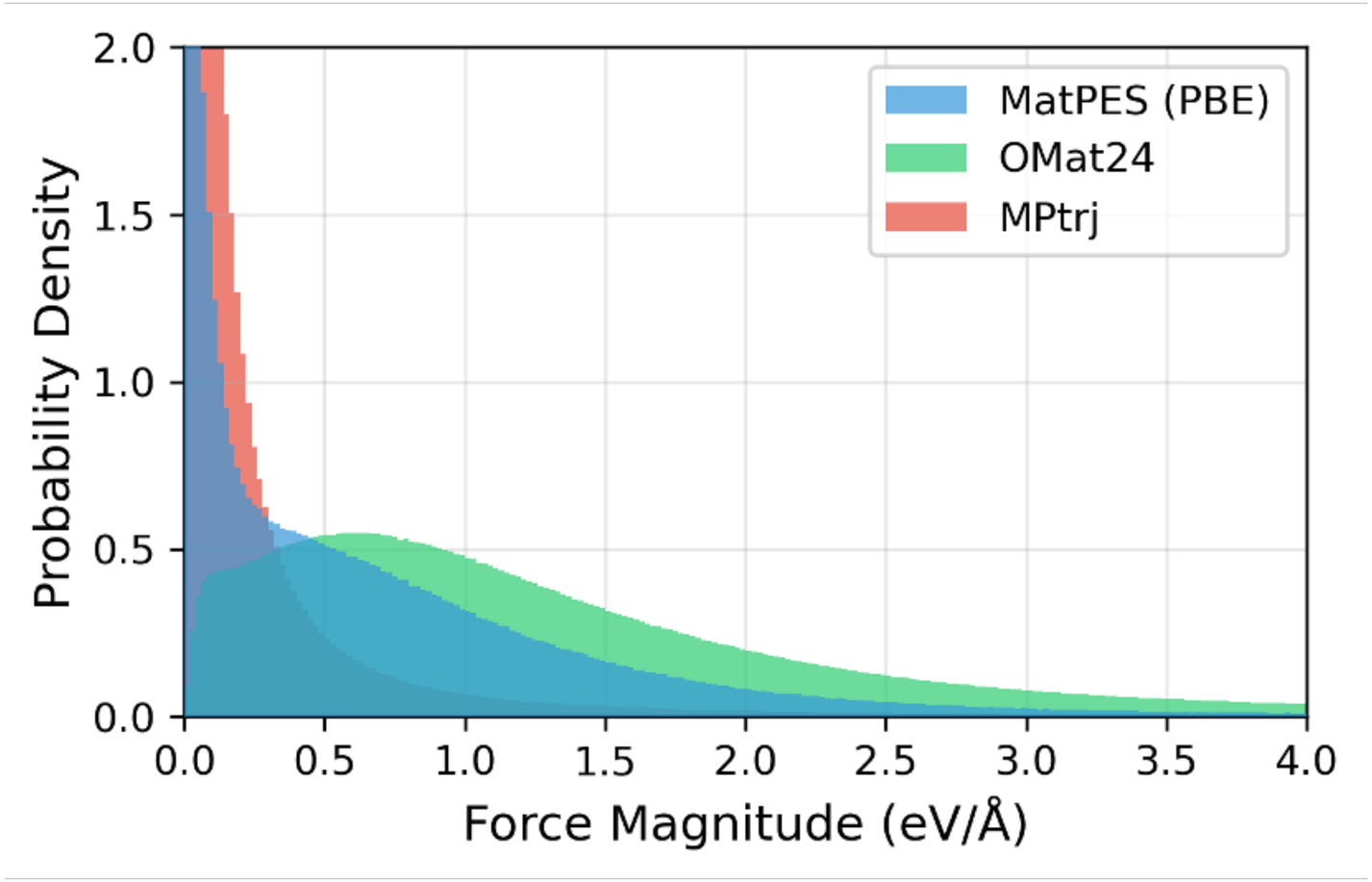
Systematic softening

- Near equilibrium training data
- Under-prediction for high-energy states
- -> limited predictive power
- -> limited application



Covering different regions of PES is important

- OMat24, MatPES datasets
- Near- and off-equilibrium configurations



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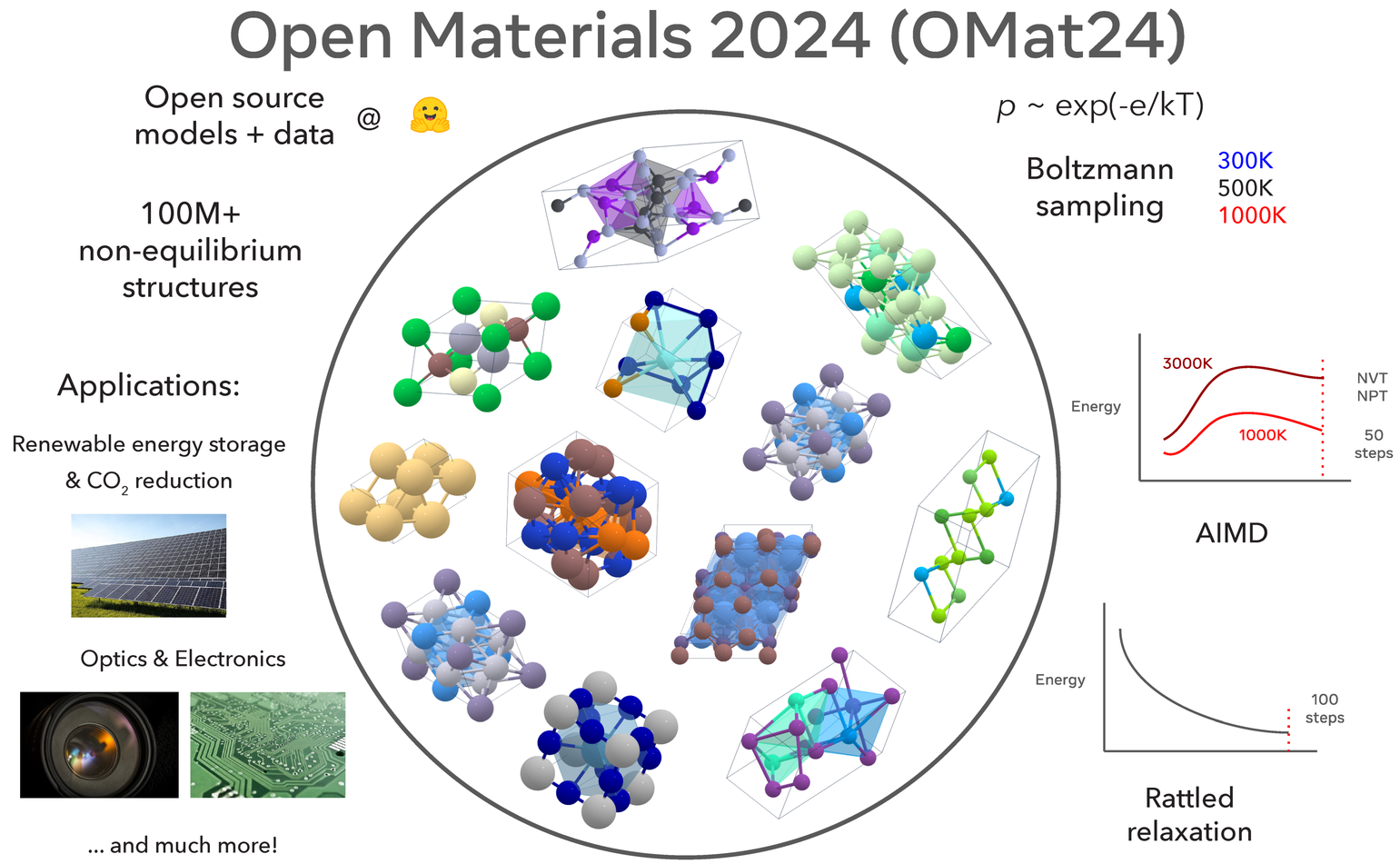
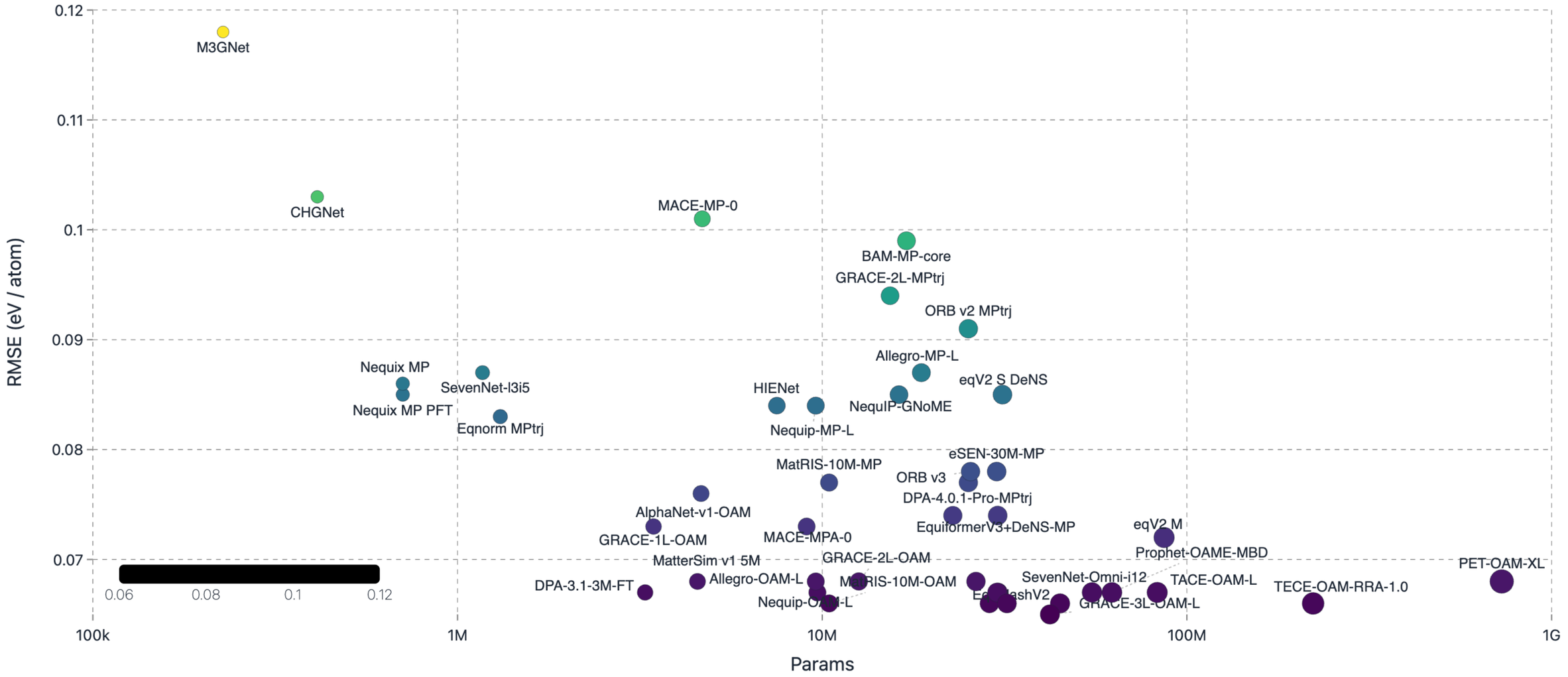


Zoo of GNN foundation potentials

- Size from 0.4M to 222M
- Mostly trained/fine-tuned on:
  - OMat24 (~101M frames)
  - sAlex (~10.5M frames)
  - MPTrj (~1.5M frames)

#	Model	MAE ↓	R^2	Params
1	TECE-OAM-RRA-1.0	0.015	0.905	222M
2	EquFlashV2	0.015	0.913	44.9M
3	EquiformerV3+DeNS-OAM	0.015	0.902	30.3M
4	PET-OAM-XL	0.015	0.906	730M
5	MatRIS-10M-OAM	0.015	0.904	10.4M
6	eqV2 M	0.015	0.904	86.6M
7	eSEN-30M-OAM	0.016	0.898	30.2M
8	EquFlash	0.017	0.902	28.7M
9	GRACE-3L-OAM-L	0.018	0.862	42.1M
10	Allegro-OAM-L	0.018	0.908	9.7M
11	TACE-OAM-L	0.019	0.867	82.9M
12	Nequip-OAM-XL	0.019	0.878	32.1M
13	SevenNet-Omni-i12*	0.019	0.868	54.9M
14	DPA-3.1-3M-FT	0.019	0.901	3.27M
15	ORB v2 MPA	0.019	0.897	25.2M
16	MatRIS-10M-MP	0.020	0.942	10.4M
17	AlphaNet-v1-OAM*	0.020	0.882	4.65M
18	Nequip-OAM-L	0.021	0.854	9.6M
19	ORB v3	0.021	0.860	25.5M
20	EquiformerV3+DeNS-MP	0.021	0.895	30.3M



https://matbench-discovery.materialsproject.org/

All in all: Universal ML interatomic potenitals

GNNs

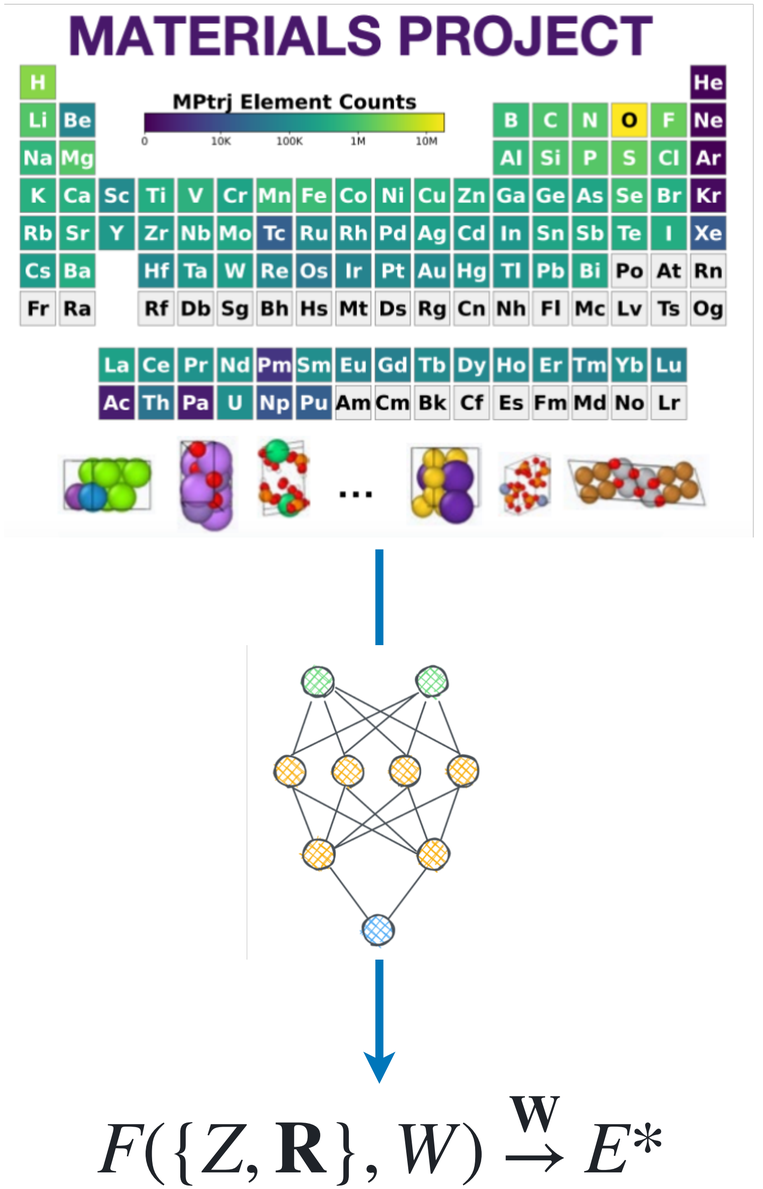
- Trained on large and diverse quantum chemistry datasets
  - The Materials Project (MPTrj), MatPES, Alexandria
- Can be applied to unknown systems
  - For solving downstream tasks

Pros:

- Works out of the box
- Can be refitted (finetuned) with a small amount of data

Cons:

- Universality is debatable. Should be carefully validated before using
- A lot of weights -> slower compared to ML potentials developed for a specific system



Take home message

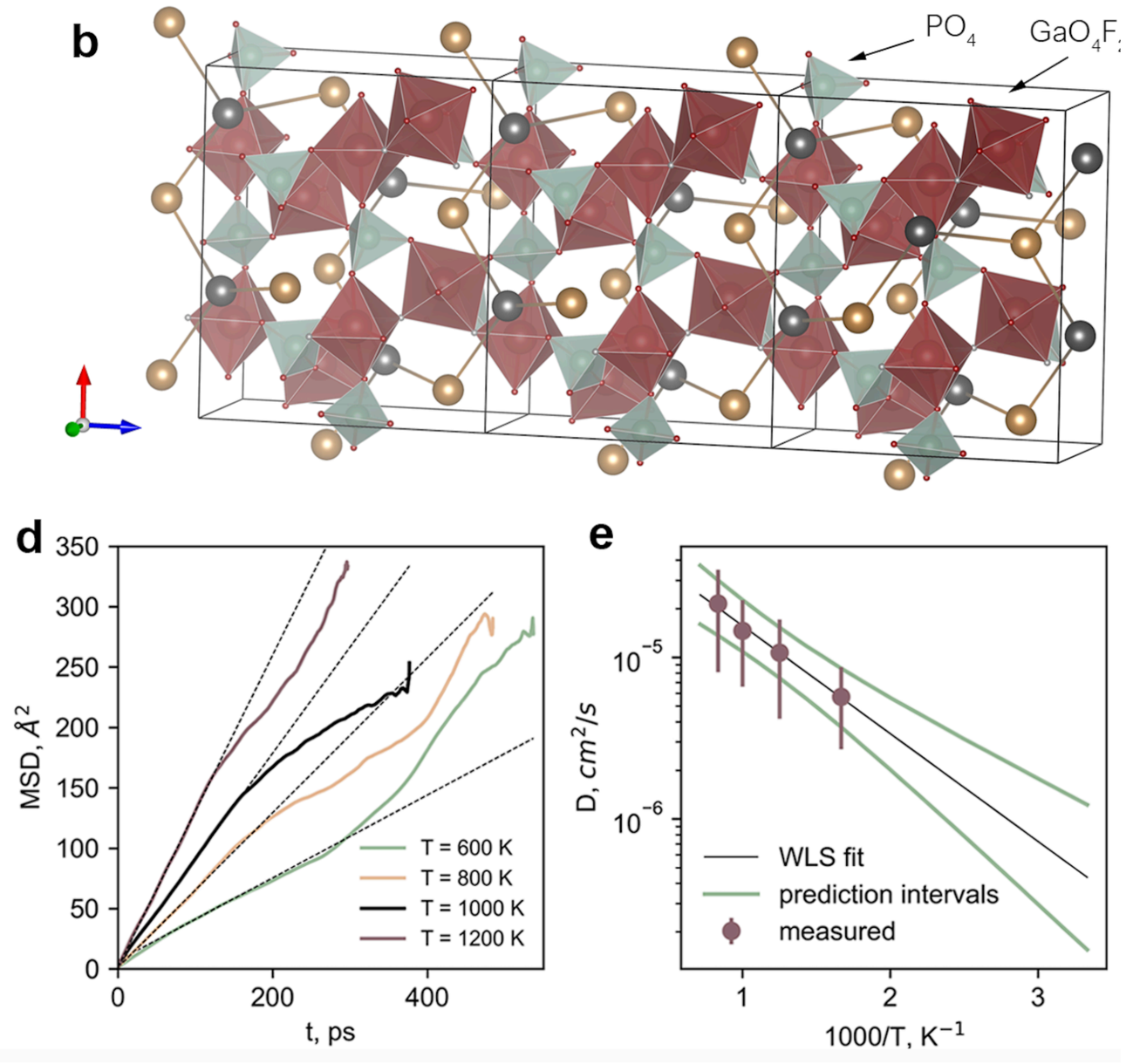
- MD is used to study the evolution of atomic systems at given conditions
- Machine learning interatomic potentials (MLIPs) are used to accelerate simulations
  - with near-DFT accuracy at a fraction of a cost
- Universal MLIPs are trained on large diverse dataset
  - to solve downstream tasks for a wide range of chemical systems
- Accuracy of universal MLIPs for a target property must be benchmarked

Ionic conductivity (for the seminar)

"is a measure of a substance's tendency towards ionic conduction.

Ionic conduction is the movement of ions.

The phenomenon is observed in solids and solutions. Ionic conduction is one mechanism of current." (wiki)



Formulas (for the seminar)

MSD vs. dt

$$MSD(\tau) = 6D\tau$$

Arrhenius equation

$$D = D_0 \exp(-E_a/kT)$$

Ionic conductivity

$\sigma = \frac{\rho e^2 D}{kT}$, where

- ρ is the diffusing particle density (number of ions N, per volume, V, of the system)
- e is the charge of electron
- D is the diffusivity
- k is Boltzman's constant